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(54) **METHOD OF USING ILOPROST FOR TREATING FROSTBITE**

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(57) **ABSTRACT**

The present disclosure generally relates to treatment of frostbite by intravenous injection or intravenous infusion of iloprost or a pharmaceutically acceptable salt thereof.

28 Claims, 4 Drawing Sheets

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Fig. 1

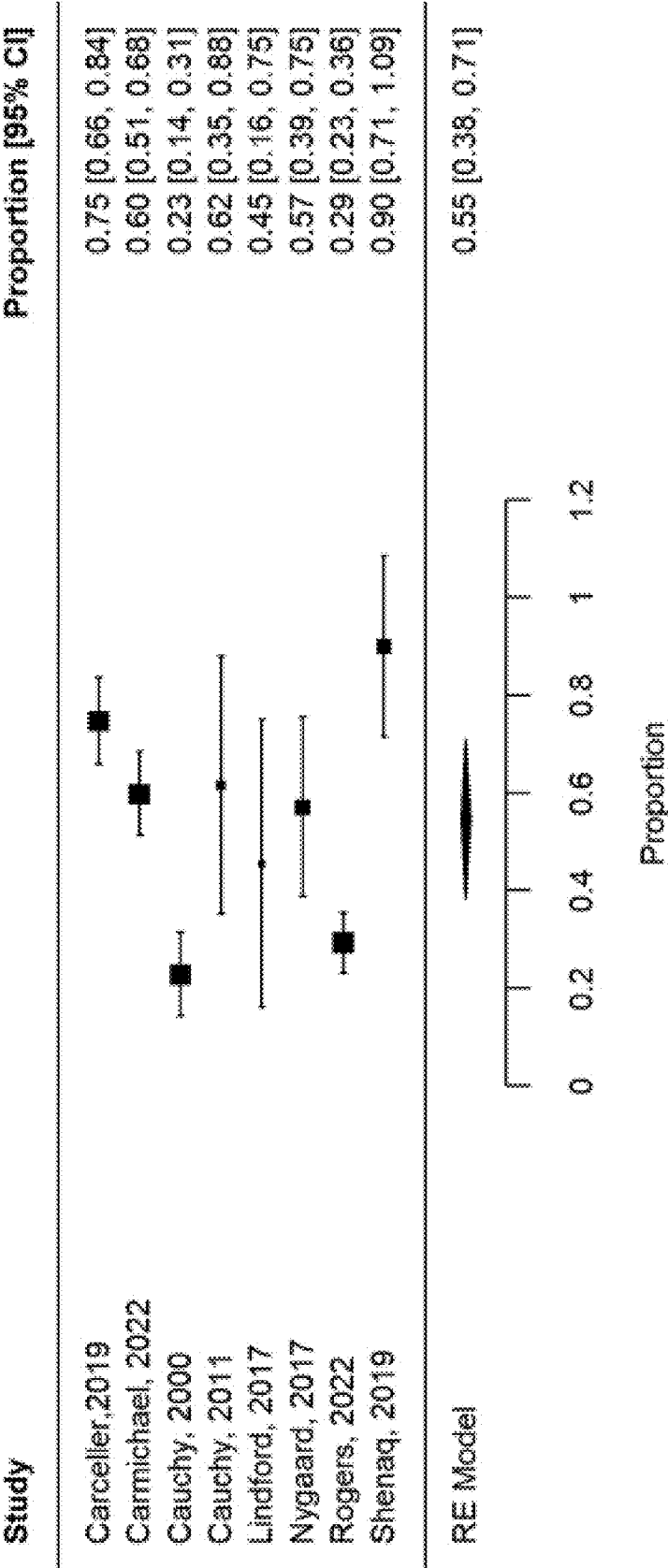


Fig. 2

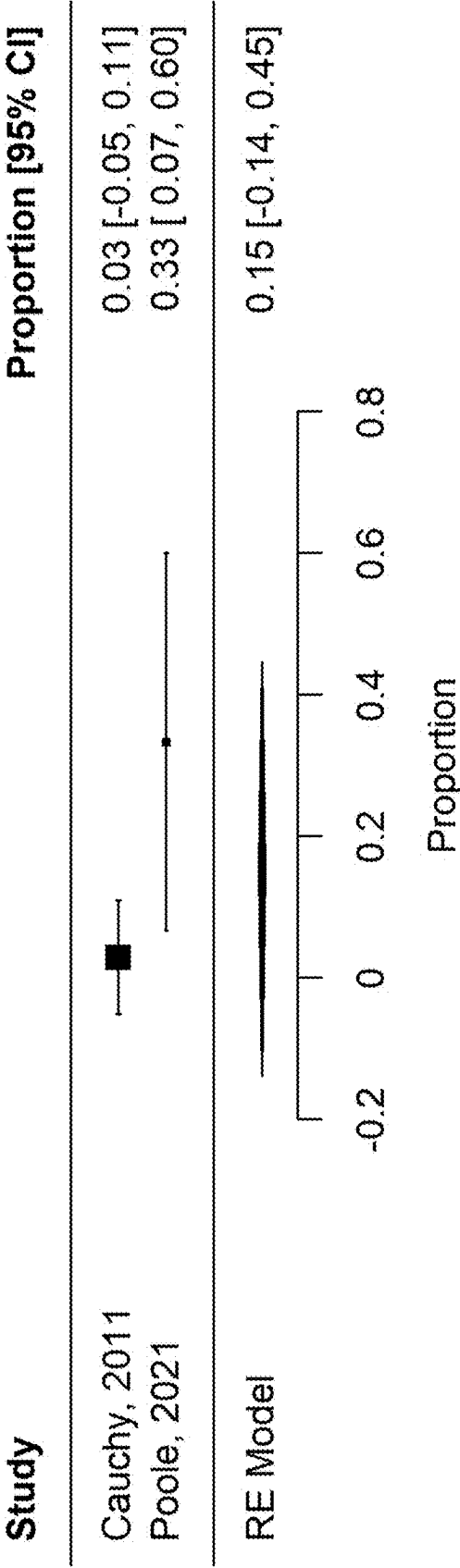


Fig. 3

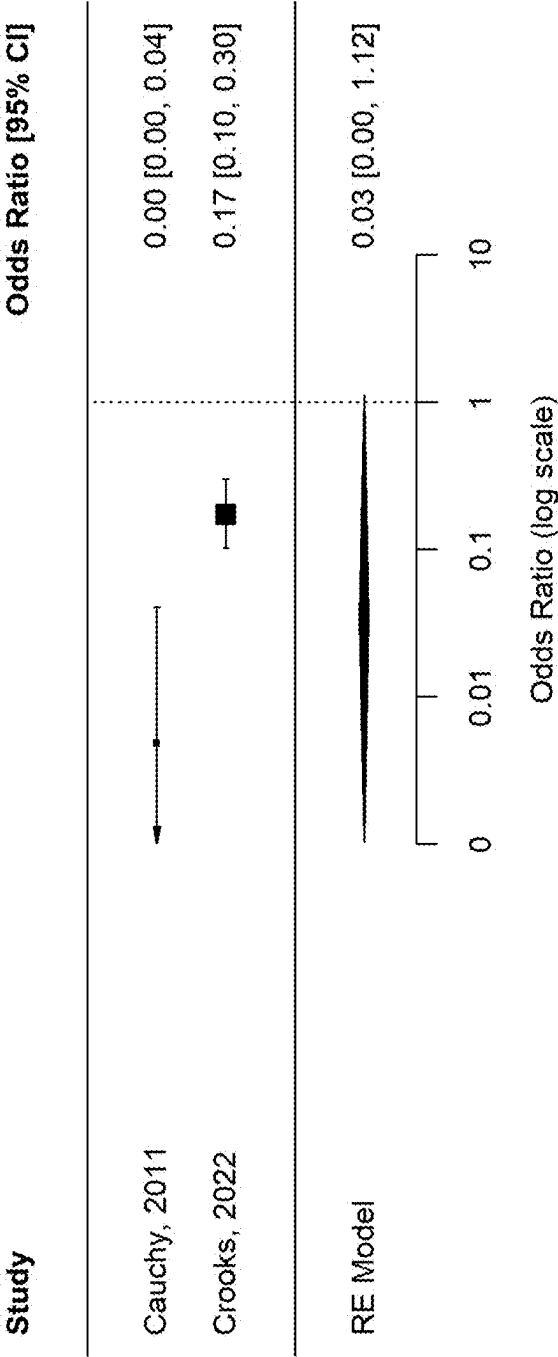
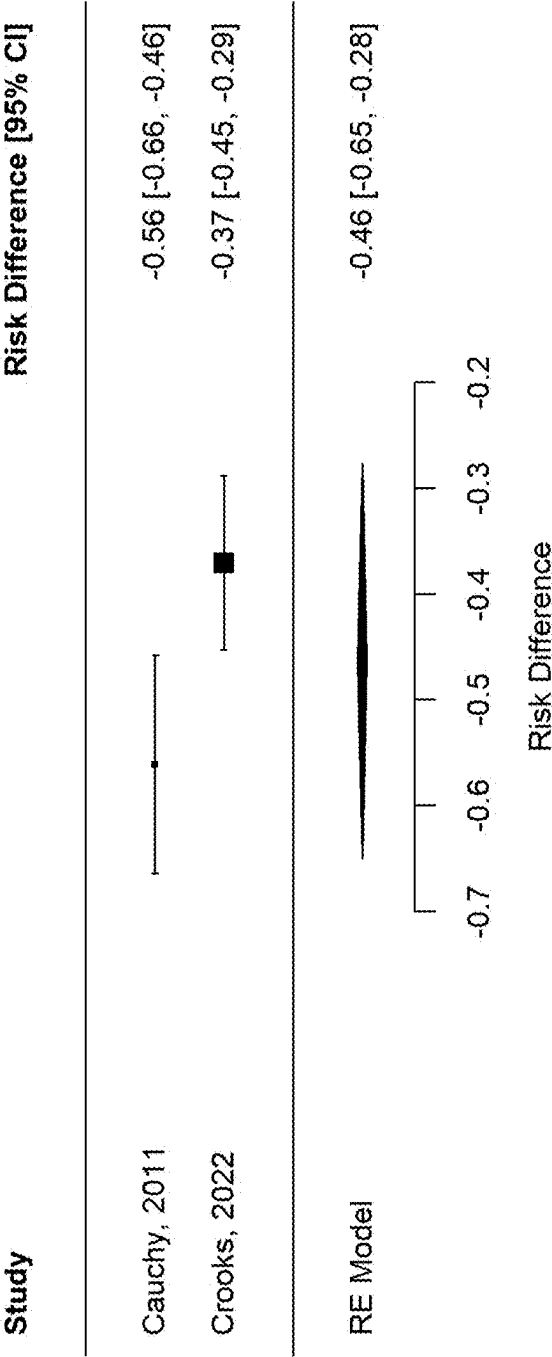


Fig. 4



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METHOD OF USING ILOPROST FOR TREATING FROSTBITE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 63/514,244, filed Jul. 18, 2023, the disclosures of which are incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

The present disclosure generally relates to treatment of frostbite by intravenous injection or intravenous infusion of iloprost or a pharmaceutically acceptable salt thereof.

BACKGROUND OF THE INVENTION

Frostbite is a thermal injury caused when tissue is exposed to freezing temperatures long enough for ice crystals to form in the affected tissue. The overall incidence of frostbite injury in the United States (US) is extremely low at 0.83 cases per year per 100,000 people. In addition, as the majority of cases are for superficial frostbite (64.8%) compared to severe frostbite injury (35.2%), severe (grade 3 and grade 4) frostbite is a very rare condition estimated at <1,000 cases annually in the US (Endorf et al., *J. Burn Care Res.* 2022, 43 (3), 646-651).

Iloprost is a stable synthetic analogue of prostacyclin (prostaglandin I₂; [PGI₂]), which exerts anti-platelet activity and reverses vasoconstriction within peripheral and systemic arterial vascular beds (Ruan et al., *Tex. Heart Inst. J.* 2010, 37 (4), 391-399; Majed et al., *Pharmacol. Rev.* 2012, 64 (3), 540-582). Iloprost infusion is associated with immediate generalized vasodilation, with the ratio of antiaggregatory:vasodilatory potency in vivo on the order of 2-7:1 (Schrör et al., *Naunyn Schmiedeberg's Arch. Pharmacol.* 1981, 316 (3), 252-255; Hermán et al., *Arch. Int. Pharmacodyn. Ther.* 1989, 300, 281-291).

Severe frostbites commonly result in amputation of the affected extremity. Traumatic digit amputations, such as amputations due to frostbite, are detrimental to the activities of daily living and work (Boulas, H. J., *J. Am. Acad. Orthop. Surg.* 1998, 6, 100-105; Sears et al., *Plast. Reconstr. Surg.* 2014, 133, 827-840). Furthermore, because the hands are so visible, digit amputation can lead to social withdrawal and reduced quality of life (Hannah, S. D. *J. Hand Ther.* 2011, 24 (2), 95-102). Thus, there is a need for an effective therapy to treat severe frostbite to prevent or reduce the risks of digit amputations.

SUMMARY OF THE INVENTION

The present disclosure provides methods of treating frostbite in a subject, comprising administering a composition comprising iloprost or a pharmaceutically acceptable salt thereof at a concentration in the range of about 0.8 mcg iloprost or a pharmaceutically acceptable salt thereof per 1 mL of pharmaceutically acceptable carrier (about 0.8 mcg/mL) to about 1.1 mcg/mL.

In one embodiment of the method as disclosed herein, the composition comprising iloprost or a pharmaceutically acceptable salt thereof has a concentration of about 0.9 mcg/mL. In some embodiments, the composition comprising iloprost or a pharmaceutically acceptable salt thereof has a concentration of about 1 mcg/mL.

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In one embodiment of the method as disclosed herein, the administering is by intravenous injection or intravenous infusion. In some embodiments, the administering is by continuous infusion.

5 In one embodiment of the method as disclosed herein, the administering is at one or more doses between about 0.2 ng/kg/min and about 3.0 ng/kg/min. In some embodiments, the administering is at one or more doses between about 0.25 ng/kg/min and about 2.0 ng/kg/min.

10 In one embodiment of the method as disclosed herein, the administering comprises a titration step on a first day of administration. In some embodiments, the administering comprises the titration step on a second day of administration. In some embodiments, the administering comprises the titration step on a third day of administration.

15 In one embodiment of the method as disclosed herein, the titration step comprises administering the composition at a starting dose of about 0.25 ng/kg/min, about 0.3 ng/kg/min, about 0.4 ng/kg/min, about 0.5 ng/kg/min, about 0.6 ng/kg/min, or about 0.7 ng/kg/min.

20 In one embodiment of the method as disclosed herein, the titration step comprises administering the composition at the starting dose of about 0.5 ng/kg/min. In some embodiments, the subject does not have a pre-existing Child-Pugh Class B or Child-Pugh Class C hepatic impairment.

25 In one embodiment of the method as disclosed herein, the dose is decreased to about 0.25 ng/kg/min at the beginning of the titration step when the subject has a renal impairment with eGFR less than 30 mL/min/m² and the subject cannot tolerate the starting dose of about 0.5 ng/kg/min.

30 In one embodiment of the method as disclosed herein, the titration step comprises administering the composition at the starting dose of about 0.25 ng/kg/min. In some embodiments, the subject has a Child-Pugh Class B or Child-Pugh Class C hepatic impairment.

35 In one embodiment of the method as disclosed herein, the titration step comprises increasing the dose every 10 minutes, 15 minutes, 20 minutes, 25 minutes, 30 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, or 60 minutes until a maximum dose is reached. In some embodiments, the dose is increased every 30 minutes.

40 In one embodiment of the method as disclosed herein, the titration step comprises increasing the dose in increments of about 0.1 ng/kg/min, about 0.2 ng/kg/min, about 0.25 ng/kg/min, about 0.3 ng/kg/min, about 0.4 ng/kg/min, about 0.5 ng/kg/min, about 0.6 ng/kg/min, about 0.7 ng/kg/min, or about 0.8 ng/kg/min until a maximum dose is reached. In some embodiments, the dose is increased in increments of about 0.5 ng/kg/min. In some embodiments, the dose is increased in increments of about 0.25 ng/kg/min.

45 In one embodiment of the method as disclosed herein, the maximum dose is predetermined. In some embodiments, the predetermined maximum dose is about 1.8 ng/kg/min, about 2.0 ng/kg/min, about 2.2 ng/kg/min, about 2.4 ng/kg/min, about 2.6 ng/kg/min, about 2.8 ng/kg/min, or about 3.0 ng/kg/min. In some embodiments, the predetermined maximum dose is about 2.0 ng/kg/min.

50 In one embodiment of the method as disclosed herein, the maximum dose is about 2.0 ng/kg/min or the highest dose the subject can tolerate between 0.25 ng/kg/min and 2.0 ng/kg/min. In some embodiments, the maximum dose is about 2.0 ng/kg/min or the highest dose the subject can tolerate between 0.5 ng/kg/min and 2.0 ng/kg/min.

55 In one embodiment of the method as disclosed herein, the titration step comprises decrease in the dose if the subject has a dose-limiting reaction. In some embodiments, the dose-limiting reaction is headache, flushing, jaw pain, myal-

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gia, nausea, or vomiting. In some embodiments, the dose is decreased in increments of about 0.5 ng/kg/min every 30 minutes until a tolerated dose is reached.

In one embodiment of the method as disclosed herein, the composition is administered continuously each day during a treatment period. In some embodiments, the continuous administration per day is about 4 hours, about 5 hours, about 6 hours, about 7 hours, or about 8 hours. In some embodiments, the continuous administration per day is about 6 hours. In some embodiments, the treatment period comprises the titration step on the first day, the second day, and/or the third day.

In one embodiment of the method as disclosed herein, once the maximum dose is determined in the titration step, the maximum dose is maintained for the remainder of the treatment period.

In one embodiment of the method as disclosed herein, the composition is administered for 4, 5, 6, 7, 8, 9, or 10 consecutive days. In some embodiments, the composition is administered for a maximum of 8 consecutive days.

In one embodiment of the method as disclosed herein, the administering is at the maximum dose found in the titration step (titration step performed on day 1, day 2, or day 3) on a fourth day and beyond.

In one embodiment of the method as disclosed herein, the administering is stopped if the subject has a dose-limiting reaction before the treatment period is over. In some embodiments, the administering is re-initiated at a previously tolerated dose after the subject's dose-limiting reaction is resolved.

In one embodiment of the method as disclosed herein, the pharmaceutically acceptable carrier is an IV fluid. In some embodiments, the pharmaceutically acceptable carrier is 0.9% sodium chloride solution, 0.45% sodium chloride solution, 0.33% sodium chloride solution, 0.225% sodium chloride solution, 5% dextrose in water (D5W), 2.5% dextrose in water (D2.5W), 5% dextrose in lactated Ringer's solution (D5LRS), Ringer's solution (RS), or sterile water. In some embodiments, the pharmaceutically acceptable carrier is 0.9% sodium chloride solution.

In one embodiment of the method as disclosed herein, the method further comprises diluting 100 mcg/mL iloprost or pharmaceutically acceptable salt thereof concentrated solution using sodium chloride solution to provide iloprost or pharmaceutically acceptable salt thereof composition with a concentration in the range of about 0.8 mcg/mL to about 1.1 mcg/mL. In some embodiments, 1 mL of the 100 mcg/mL iloprost or pharmaceutically acceptable salt thereof concentrated solution is in a single dose vial. In some embodiments, the single dose vial is a glass vial.

In one embodiment of the method as disclosed herein, the most common adverse event is headache, flushing, palpitations/tachycardia, nausea, vomiting, dizziness, or hypotension.

In one embodiment of the method as disclosed herein, safety profile of the method in the subject is consistent with safety profile observed in systemic sclerosis subjects experiencing symptomatic digital ischemic episodes from multicenter, double-blind, randomized, placebo-controlled studies with iloprost continuous infusion.

In one embodiment of the method as disclosed herein, the frostbite is a stage 3 or a stage 4 frostbite. In some embodiments, the frostbite is a severe frostbite.

In one embodiment of the method as disclosed herein, the method prevents or reduces the risk of digit amputation.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows proportion of patients with amputation after standard of care treatment for frostbite.

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FIG. 2 shows proportion of patients with amputation after iloprost treatment for frostbite.

FIG. 3 shows digits amputated following treatment with iloprost alone vs standard of care.

FIG. 4 shows digits amputated following treatment with iloprost alone and iloprost plus tPA vs standard of care.

DETAILED DESCRIPTION

All publications, patents and patent applications, including any drawings and appendices therein are incorporated by reference in their entirety for all purposes to the same extent as if each individual publication, patent or patent application, drawing, or appendix was specifically and individually indicated to be incorporated by reference in its entirety for all purposes.

Definitions

While the following terms are believed to be well understood by one of ordinary skill in the art, the following definitions are set forth to facilitate explanation of the presently disclosed subject matter.

Throughout the present specification, the terms "about" and/or "approximately" may be used in conjunction with numerical values and/or ranges. The term "about" is understood to mean those values near to a recited value. Furthermore, the phrases "less than about [a value]" or "greater than about [a value]" should be understood in view of the definition of the term "about" provided herein. The terms "about" and "approximately" may be used interchangeably.

Throughout the present specification, numerical ranges are provided for certain quantities. It is to be understood that these ranges comprise all subranges therein. Thus, the range "from 50 to 80" includes all possible ranges therein (e.g., 51-79, 52-78, 53-77, 54-76, 55-75, 60-70, etc.). Furthermore, all values within a given range may be an endpoint for the range encompassed thereby (e.g., the range 50-80 includes the ranges with endpoints such as 55-80, 50-75, etc.).

The term "a" or "an" refers to one or more of that entity; for example, "a therapeutic agent" refers to one or more therapeutic agents or at least one therapeutic agent. As such, the terms "a" (or "an"), "one or more" and "at least one" are used interchangeably herein. In addition, reference to "an inhibitor" by the indefinite article "a" or "an" does not exclude the possibility that more than one of the inhibitors is present, unless the context clearly requires that there is one and only one of the inhibitors.

As used herein, the verb "comprise" as is used in this description and in the claims and its conjugations are used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. The present invention may suitably "comprise", "consist of", or "consist essentially of", the steps, elements, and/or reagents described in the claims.

It is further noted that the claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as "solely", "only" and the like in connection with the recitation of claim elements, or the use of a "negative" limitation.

The term "pharmaceutically acceptable salts" includes both acid and base addition salts. Pharmaceutically acceptable salts include those obtained by reacting the active compound functioning as an acid, with an inorganic or organic base to form a salt. Organic base includes, but are

not limited to, monoethanolamine, diethanolamine, triethanolamine, trometamol and meglumine. Those skilled in the art will further recognize that base addition salts may be prepared by reaction of the compounds with the appropriate inorganic or organic base via any of a number of known methods.

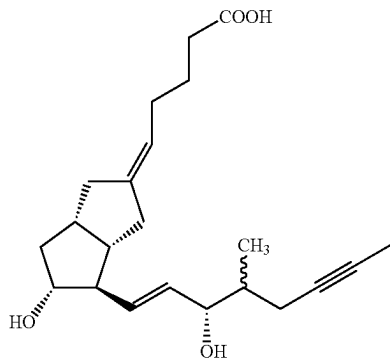
The term “treating” means one or more of relieving, alleviating, delaying, reducing, improving, or managing at least one symptom of a condition in a subject. The term “treating” may also mean one or more of arresting, delaying the onset (i.e., the period prior to clinical manifestation of the condition) or reducing the risk of developing or worsening a condition.

The compound of the invention, or their pharmaceutically acceptable salts contain asymmetric centers and can thus give rise to enantiomers, diastereomers, and other stereoisomeric forms that can be defined, in terms of absolute stereochemistry, as (R) or (S). The present disclosure is meant to include all such possible isomers, as well as their racemic and optically pure forms whether or not they are specifically depicted herein. Stereoisomers can be prepared using chiral synthons or chiral reagents, or resolved using conventional techniques, for example, chromatography and fractional crystallization. Conventional techniques for the preparation/isolation of individual enantiomers include chiral synthesis from a suitable optically pure precursor or resolution of the racemate (or the racemate of a salt or derivative) using, for example, chiral high pressure liquid chromatography (HPLC). The compound described herein also contains an olefinic double bond, and unless specified otherwise, it is intended that the compounds include both E and Z geometric isomers.

A “stereoisomer” refers to a compound made up of the same atoms bonded by the same bonds but having different three-dimensional structures, which are not interchangeable. The present disclosure contemplates various stereoisomers and mixtures thereof and includes diastereomers.

Pharmaceutically Active Ingredient

The present disclosure relates to use of iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof for treating frostbite. Iloprost has the following structure and can also be identified as (5E)-5-[(3aS,4R,5R,6aS)-5-hydroxy-4-[(E,3S)-3-hydroxy-4-methyloct-1-en-6-ynyl]-3,3a,4,5,6,6a-hexahydro-1H-pentalen-2-ylidene]pentanoic acid.



In some embodiment, iloprost consists of a mixture of the 4R and 4S diastereomers. In some embodiments, iloprost consists of a mixture of the 4R and 4S diastereomers at a

ratio of about 1:1. In some embodiments, iloprost consists of a mixture of the 4R and 4S diastereomers at a ratio of about 53:47.

The present disclosure also relates to pharmaceutical composition comprising iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof.

International Application Publication Nos. WO222/032141 and WO2022/036234 disclosed iloprost composition and treatment, each of which is hereby incorporated by reference in its entirety for all purposes.

Iloprost Composition

The present disclosure relates to administering a composition comprising iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof. In some embodiment, the composition is a pharmaceutical composition.

In some embodiments, iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof, is formulated for injection. In some embodiments, iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof, is formulated for intravenous injection or intravenous infusion. In some embodiments, iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof, is formulated for continuous infusion.

In embodiments, iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, is formulated as a sterile solution. In embodiments, a composition comprising iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof, further comprises a pharmaceutically acceptable carrier or excipient.

Pharmaceutically acceptable excipients include, but are not limited to, solubilizing agents, pH adjusting agents, tonicity agents, buffering agents, and/or solvents.

In embodiments, solubilizing agents is selected from pharmaceutically acceptable alcohols, glycols, esters, ethers, or silicones. In embodiments, the solubilizing agent is ethanol.

In embodiments, pH adjusting agents is a pharmaceutically acceptable acid or base. In embodiments, the pH adjusting agent is hydrochloric acid.

In embodiments, tonicity agents include, but are not limited to, glycerin, lactose, mannitol, dextrose, sodium chloride, sodium sulfate, and sorbitol. In embodiments, the tonicity agent is sodium chloride.

In embodiments, buffering agents include, but not limited to, citrate buffer, phosphate buffer, phosphate citrate buffer, bicarbonate buffer, tartrate buffer, acetate buffer, and trometamol buffer. In embodiments, the buffering agent is trometamol.

In embodiments, solvents are sterile solvents. In some embodiments, the sterile solvent is an IV fluid. In embodiments, solvents include, but are not limited to, water, glucose solution, dextrose solution, saline solution, Ringer's solution, and lactated Ringer's solution. In some embodiments, saline solution is 0.9% sodium chloride solution, 0.45% sodium chloride solution, 0.33% sodium chloride solution, or 0.225% sodium chloride solution. In some embodiments, dextrose solution is 5% dextrose in water (D5W), 2.5% dextrose in water (D2.5W) or 5% dextrose in lactated Ringer's solution (DSLRS). In embodiments, the solvent is water. In some embodiments, the solvent is 0.9% sodium chloride solution.

In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, further comprises trometamol, ethanol, sodium chloride, hydrochloric acid, and water.

In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, contains no preservatives.

In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, is provided as a single use vial. In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, is provided as a single dose vial. In some embodiments, the single use vial or the single dose vial is a glass vial. In embodiments, each vial comprises about 100 mg iloprost. In embodiments, each vial comprises about 100 mcg iloprost per 1 mL of pharmaceutically acceptable carrier and/or excipients. In embodiments, each single use vial or single dose vial has composition comprising about 100 mcg iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof. In embodiments, each single use vial or single dose vial has a composition comprising about 100 mcg (0.1 mg) iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof, about 8.1 mg ethanol, about 0.242 mg tromethamine, and about 0.9 mg sodium chloride. In embodiments, each single use vial or single dose vial has a pH of about 8.3. In embodiments, each single use vial or single dose vial has a pH of about 8.3 which is adjusted with hydrochloric acid and sodium hydroxide. In embodiments, each single use vial or single dose vial has a volume of 1 mL which is filled with water for injection to make 1 mL after above-described ingredients are measured.

A glass vial comprising 100 mcg iloprost per 1 mL of pharmaceutically acceptable carrier and/or excipients provides an important advancement in the pharmaceutical use of iloprost.

Before approval of AURLUMYN™, all approved iloprost was sold in glass ampules around the world. Glass ampules are associated with safety risks including percutaneous injuries to healthcare professionals when breaking glass ampules and accidental transfer of glass particulate matter during pharmacy compounding. Accidental transfer of glass particulate, which could be a result of health professional not using an appropriate filter needle when transferring the ampule contents, is especially dangerous as it can result in the injection of glass particles to patients and cause serious adverse events. Adverse events associated with accidental injection of glass particulates include pulmonary thrombi, micro-emboli, infusion phlebitis, end-organ granuloma formation and inflammation. These risks associated with the use of glass ampules can be eliminated by providing the therapeutics in a glass vial. In addition, a single use vial would further reduce potential mistakes in diluting the therapeutics for infusion compared to needing to combine the contents of two or more drug preparations (ampule or vial) or syringing out a set amount from an appropriate vessel containing larger quantities of the therapeutics.

In embodiments, the single use or the single dose vial containing iloprost is stored at room temperature (20° C. to 25° C.). In embodiments, the single use or the single dose vial containing iloprost is stored at a temperature between about 20° C. to about 25° C. with allowed temperature excursions permitted to 15° C. to 30° C. In embodiments, the single use or the single dose vial containing iloprost is stored at a temperature between about 20° C. to about 25° C. In embodiments, the single use or the single dose vial containing iloprost is stored at a refrigerated temperature (approximately 5° C.). In embodiments, the single use or the single dose vial containing iloprost is protected from light when stored. In embodiments, the single use or the single dose vial contains 100 µg iloprost per 1 mL.

In embodiments, the single use or the single dose vial containing iloprost is stable at 25° C. for at least 6 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 25° C. for at least 12 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 25° C. for at least 18 months.

In embodiments, the single use or the single dose vial containing iloprost is stable at 25° C. and at 60% relative humidity (RH) for at least 6 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 25° C./60% RH for at least 12 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 25° C./60% RH for at least 18 months.

In embodiments, the single use or the single dose vial containing iloprost is stable at 40° C. for at least 1 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 40° C. for at least 3 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 40° C. for at least 6 months.

In embodiments, the single use or the single dose vial containing iloprost is stable at 40° C. and at 75% relative humidity (RH) for at least 1 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 40° C./75% RH for at least 3 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 40° C./75% RH for at least 6 months.

In embodiments, the single use or the single dose vial containing iloprost is stable at 5° C. for at least 3 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 5° C. for at least 6 months. In embodiments, the single use or the single dose vial containing iloprost is stable at 5° C. for at least 9 months.

In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, has a pH of about 7.0 to about 9.0. In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, has a pH of about 8.0 to about 9.0.

In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, is further diluted with a sterile solvent for injection or infusion. In embodiments, the composition is further diluted with 0.9% sodium chloride solution, 0.45% sodium chloride solution, 0.33% sodium chloride solution, 0.225% sodium chloride solution, 5% dextrose in water (D5W), 2.5% dextrose in water (D2.5W), 5% dextrose in lactated Ringer's solution (D5LRS), Ringer's solution (RS), or sterile water. In embodiments, the composition is further diluted with saline solution for injection or infusion. In some embodiments, the composition is further diluted with 0.9% sodium chloride solution. In embodiments, the composition is further diluted with 0.9% sodium chloride injection, USP.

In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, for injection or infusion is at a concentration in the range of about 0.8 mcg iloprost or a pharmaceutically acceptable salt thereof per 1 mL of pharmaceutically acceptable carrier (0.8 mcg/mL) to about 1.1 mcg/mL. In some embodiments, the composition or the composition has a concentration of about 0.9 mcg/mL. In some embodiments, the composition or the composition has a concentration of about 1 mcg/mL (1,000 ng/mL).

In embodiments, 1 mL of the composition comprising 100 mcg of iloprost or pharmaceutically acceptable salt thereof per 1 mL is diluted with 99 mL of 0.9% sodium chloride. In embodiments, 1 mL of 100 mcg iloprost/mL is diluted with

99 mL of 0.9% sodium chloride injection, USP, to provide iloprost concentration of 1,000 ng/mL (1 mcg/mL).

In embodiments, 1 mL of the composition comprising 100 mcg of iloprost or pharmaceutically acceptable salt thereof per 1 mL is added to an infusion bag labeled to contain 100 mL 0.9% sodium chloride injection, USP, without concern for overfill. In embodiments, 1 mL of 100 mcg iloprost/mL is diluted with 0.9% sodium chloride injection, USP, to provide iloprost concentration between about 0.8 mcg/mL and about 1.1 mcg/mL.

In embodiments, diluted composition is ready to use and can be used immediately, or stored at refrigerated temperatures (2° C. to 8° C.) for a maximum of 8 days prior to use, or stored at room temperature (20° C. to 25° C.) for 4 hours prior to administration as a 6-hour continuous infusion. In embodiments, continuous infusion is continuous intravenous infusion.

In embodiments, a composition comprising iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, for injection or infusion is at a concentration of about 25 µg iloprost per 1 mL. In embodiments, 25 µg iloprost per 1 mL composition is ready to use and can be used immediately, or stored at refrigerated temperatures (2° C. to 8° C.) for a maximum of 8 days prior to use, or stored at room temperature (20° C. to 25° C.) for 4 hours prior to administration as a 6-hour continuous infusion. In embodiments, continuous infusion is continuous subcutaneous infusion.

In embodiments, ready to use composition comprising iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof, as disclosed herein is safe, efficacious, and stable for up to 8 days at 2° C. to 8° C. The 8-day stability for the ready to use composition is critical so that patients receive an accurate dose of iloprost or a pharmaceutically acceptable salt or a stereoisomer thereof and to minimize the risk of infection (sepsis or line-infection). In embodiments, the 8-day stability for the ready to use composition is important in enabling decentralized infusions in the patient's home setting or at an off-site ambulatory infusion sites, which reduces the patients' risk to nosocomial infections (contracting infection at a hospital or infusion center) and will improve patient convenience which reduces the risk of non-compliance.

In embodiments, a subcutaneous continuous infusion would use a more concentrated fully diluted ready to use iloprost product (e.g., 25 µg/mL instead of 1 µg/mL for intravenous infusion) but the dose delivered (ng/kg/min) and time of delivery (6 hours) would be equivalent.

Therapeutic Use

The present disclosure relates to administering iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof to a frostbite patient. In embodiments, the iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof is administered by injection. In embodiments, administration is by intravenous or subcutaneous injection. In embodiments, administration is by intravenous injection or intravenous infusion. In embodiments, administration is by continuous infusion. In embodiments, administration is by continuous infusion is intravenous infusion or subcutaneous infusion. In embodiments, administration is through peripheral catheter system, a peripheral inserted central catheter (PICC), or subcutaneous catheter in the abdomen. In embodiments, administration is through NovaCath Integrated IV Catheter System or a Poly Per-Q-Cath Catheter. In embodiments, the same peripheral catheter system or a peripheral inserted central catheter (PICC) is used for 1, 2, 3, 4, 5, 6, 7, or 8 days of treatment. In embodiments, the

same peripheral catheter system or a peripheral inserted central catheter (PICC) is used during all consecutive days of treatment.

In embodiments of administering iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof to a frostbite patient, the administration is done at a medical facility by a medically trained professional. In embodiments, the administration is done at a decentralized setting. In embodiments, the administration is done at the subject's home or ambulatory infusion suite. In embodiments of the administration at a decentralized setting, the administration of iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof is performed by a medically trained professional. In embodiments of the administration at a decentralized setting, a physician is accessible by telehealth during the treatment to assess adverse events and vital signs.

In some embodiments, iloprost or a pharmaceutically acceptable salt thereof is administered to a subject at a concentration in the range of about 0.8 mcg iloprost or a pharmaceutically acceptable salt thereof per 1 mL of pharmaceutically acceptable carrier (about 0.8 mcg/mL) to about 1.1 mcg/mL. In some embodiments, iloprost or a pharmaceutically acceptable salt thereof is administered to a subject at a concentration of about 0.8 mcg/mL, about 0.9 mcg/mL, about 1.0 mcg/mL, or about 1.1 mcg/mL.

In some embodiments, concentrated solution of iloprost or a pharmaceutically acceptable salt thereof is diluted prior to therapeutic use. In some embodiments, the dilution is with a suitable IV fluid or suitable sterile injection solution. In some embodiments, the dilution is with 0.9% Sodium Chloride Injection, USP.

In some embodiments, 1 mL of the composition comprising 100 mcg of iloprost or pharmaceutically acceptable salt thereof per 1 mL is diluted with 99 mL of 0.9% sodium chloride solution. In embodiments, 1 mL of 100 mcg iloprost/mL is transferred into a 100 mL 0.9% sodium chloride solution infusion bag. In embodiments, 1 mL of 100 mcg iloprost/mL is transferred into a commercially available infusion bag labeled to contain 100 mL 0.9% sodium chloride solution (which contains overfill so the actual volume will be larger than 100 mL).

The present disclosure also relates to method for treatment of frostbite. In some embodiments, frostbite is severe frostbite. In some embodiments, frostbite is grade 3 or grade 4 frostbite.

In some embodiment, frostbite subject has at least one digit (finger or toe) affected by frostbite. In some embodiment, frostbite subject has at least one digit (finger or toe) with frostbite stage 3 (lesion extending just past the proximal phalanx) or stage 4 (lesion extending proximal to the metacarpal or metatarsal joint).

The present disclosure also relates to method for treatment of frostbite to reduce the risk of digit amputations.

The present disclosure also relates to administering iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof to a frostbite subject by intravenous injection, subcutaneous injection, intravenous infusion, or subcutaneous infusion. In embodiments, administration is by intravenous infusion or intravenous injection. In embodiments, administration is by continuous intravenous infusion.

In embodiments, an advantage of continuous infusion of iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof, is its greater bioavailability when compared to other routes of administration. For example, iloprost (a synthetic analog of prostacyclin, PGI₂) has poor oral bioavailability and tolerability making oral administration route not viable.

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In embodiments, another advantage of continuous infusion of iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof, is that the iloprost composition disclosed herein for infusion is stable which allows for home infusion and decentralized infusions.

In embodiments, another advantage of continuous infusion of iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof, is the ability for iloprost to act as a potent prostacyclin (IP-) receptor agonist. In embodiments, iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof, increases cyclic AMP concentrations in pertinent cells thereby having an effect as vasodilator or as an anti-vasoconstrictor, anti-fibrotic, anti-platelet, and/or anti-inflammatory. In embodiments, once iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof, reaches the site of action (digit cutaneous circulation), the infusion therapy would inhibit platelet aggregation.

As used herein, a treatment period refers to the duration to which a subject received iloprost or a pharmaceutically acceptable salt or stereoisomer thereof for the treatment for frostbite. In embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered once a day for 3 to 10 consecutive days (i.e., the treatment period is 3 to 10 days). In embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered once a day for 3 to 8 consecutive days. In embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered once a day for up to 8 consecutive days.

In embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered once a day via intravenous injection or infusion over about 4 hours to about 8 hours. In embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered once a day via intravenous injection or infusion over about 4 hours, about 5 hours, about 6 hours, about 7 hours, or about 8 hours. In embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered once a day via intravenous injection or infusion over about 6 hours.

In some embodiments, the administration is continuous, unless there is a need to pause the administration due to adverse events. In some embodiments, the continuous administration of iloprost or a pharmaceutically acceptable salt or stereoisomer thereof per day is for about 4 hours, about 5 hours, about 6 hours, about 7 hours, or about 8 hours. In some embodiments, the continuous administration of iloprost or a pharmaceutically acceptable salt or stereoisomer thereof per day is for about 6 hours. In some embodiments, the continuous administration of iloprost or a pharmaceutically acceptable salt or stereoisomer thereof per day is for about 6 hours and the treatment period is a maximum of 8 consecutive days. In some embodiments, the continuous administration is via intravenous injection or infusion.

In some embodiments, the continuous administration of iloprost or a pharmaceutically acceptable salt or stereoisomer thereof is once a day for about 4 hours, about 5 hours, about 6 hours, about 7 hours, or about 8 hours. In some embodiments, the continuous administration of iloprost or a pharmaceutically acceptable salt or stereoisomer thereof is once a day for about 6 hours. In some embodiments, the continuous administration of iloprost or a pharmaceutically acceptable salt or stereoisomer thereof is once a day for about 6 hours and the treatment period is a maximum of 8 consecutive days.

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In some embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered once a day via continuous infusion for 6 hours. In some embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered once a day via continuous infusion for 6 hours for a maximum of 8 consecutive days.

In embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered at a rate or a dose in the range of about 0.2 ng iloprost/kg body weight/min (ng/kg/min) to about 3.0 ng/kg/min. In embodiments, iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is administered at a rate or a dose in the range of about 0.25 ng/kg/min to about 2.0 ng/kg/min. In embodiments, the rate or the dose of injection or infusion is adjusted according to the subject's tolerability within the range of 0.25 to about 2.0 ng/kg/min. In embodiments, the rate or the dose of injection or infusion is adjusted according to the subject's tolerability within the range of 0.5 to about 2.0 ng/kg/min.

In embodiments, the administration rate or dose is titrated on the first day of treatment. In some embodiments, titration step is repeated on the second day of treatment. In some embodiments, titration step is repeated on the third day of treatment. In some embodiments, the treatment on the fourth day starts with the highest tolerated dose from the third day. During the iloprost treatment, the dose or the rate may be adjusted based on tolerability of the subject.

In embodiments, on the first day of treatment, administration of iloprost or a pharmaceutically acceptable salt or stereoisomer thereof, is initiated at a rate or a dose of about 0.25 ng/kg/min, about 0.3 ng/kg/min, about 0.4 ng/kg/min, about 0.5 ng/kg/min, about 0.6 ng/kg/min, or about 0.7 ng/kg/min (starting dose). In some embodiments, the starting dose is 0.5 ng/kg/min.

In some embodiments, the starting dose is 0.5 ng/kg/min if the subject does not have a pre-existing Child-Pugh Class B or Child-Pugh Class C hepatic impairment. In some embodiments, the starting dose is 0.5 ng/kg/min but the dose is decreased to about 0.25 ng/kg/min at the beginning of the titration step when the subject has a renal impairment with eGFR less than 30 mL/min/m² and the subject cannot tolerate the starting dose of about 0.5 ng/kg/min.

In some embodiments, the starting dose is 0.25 ng/kg/min. In some embodiments, the starting dose is 0.25 ng/kg/min if the subject has a Child-Pugh Class B or Child-Pugh Class C hepatic impairment.

In embodiments, the rate or the dose of administration is increased stepwise during titration every 10 minutes, 15 minutes, 20 minutes, 25 minutes, 30 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, or 60 minutes until a maximum dose is reached. In some embodiments, the rate or the dose of administration is increased stepwise during titration every 30 minutes until a maximum dose is reached.

In embodiments, the rate or the dose of administration is increased stepwise during titration in increments of about 0.1 ng/kg/min, about 0.2 ng/kg/min, about 0.25 ng/kg/min, about 0.3 ng/kg/min, about 0.4 ng/kg/min, about 0.5 ng/kg/min, about 0.6 ng/kg/min, about 0.7 ng/kg/min, or about 0.8 ng/kg/min until a maximum dose is reached. In some embodiments, the rate or the dose of administration is increased stepwise during titration in increments of about 0.5 ng/kg/min. In some embodiments, the rate or the dose of administration is increased stepwise during titration in increments of about 0.25 ng/kg/min.

In some embodiments, the maximum dose is predetermined. In some embodiments, the predetermined maximum dose is about 1.8 ng/kg/min, about 2.0 ng/kg/min, about 2.2

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ng/kg/min, about 2.4 ng/kg/min, about 2.6 ng/kg/min, about 2.8 ng/kg/min, or about 3.0 ng/kg/min. In some embodiments, the predetermined maximum dose is about 2.0 ng/kg/min.

In some embodiments, the maximum dose is about 2.0 ng/kg/min or the highest dose the subject can tolerate between 0.25 ng/kg/min and 2.0 ng/kg/min.

In some embodiments, once the maximum dose is reached on a titration day (i.e., day 1, day 2, and/or day 3), the maximum dose is maintained for the remainder of the treatment for that day.

In embodiments, the rate or the dose of administration is increased about every 30 minutes in increments of 0.5 ng/kg/min up to 2.0 ng/kg/min to determine the maximum dose on the first day of the treatment. In embodiments, the second day and the third day of the treatment repeats the titration steps of the first day. In embodiments, the fourth day and thereafter is initiated at the maximum dose tolerated by the subject on treatment day 3. In embodiments, the rate or the dose of injection or infusion can be adjusted by the administering medical professional at any time during the treatment.

In some embodiments, the titration comprises decreasing in the dose if the subject has a dose-limiting reaction. In some embodiments, the dose-limiting reaction is headache, flushing, jaw pain, myalgia, nausea, or vomiting.

In some embodiments, the titration comprises decreasing in the dose if the subject has an adverse event. In some embodiments, the adverse event is headache, flushing, palpitations/tachycardia, nausea, vomiting, dizziness, or hypotension.

In embodiments, if the administration of iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof is stopped due to a dose-limiting adverse event, administration can be reinitiated at a previously tolerated rate or dose once the adverse event or the dose-limiting reaction is resolved.

In embodiments of administering iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof to a frostbite patient, the administration is temporarily interrupted if the patient experiences symptomatic hypotension, systolic blood pressure <80 mm HG, intolerable adverse events (e.g., vomiting), or a systolic blood pressure drop more than 10 mm Hg from the patient's pre-infusion measurement. In embodiments, when the patient's systolic blood pressure drops more than 10 mm Hg from the patient's pre-infusion measurement, a physician must determine if the infusion should be re-initiated after correcting the hypotension. In embodiments, symptomatic hypotension is any reduction of blood pressure associated with symptoms (e.g., dizziness, lightheadedness, syncope). In embodiments where the treatment is temporarily interrupted, a physician determines whether to re-initiate treatment after correcting hypotension.

In embodiments, iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof is administered to a frostbite subject through a peripheral line or peripherally inserted central catheter using an infusion pump. In embodiments of infusion administration, in-line 0.22 micron filter or 0.2 micron filter is used. In embodiments of infusion administration, infusion pump should be able to deliver rates between 0.1 to 99.99 mL per hour. In embodiments of infusion administration, infusion pump should be able to adjust infusion rates with increments of 0.1 mL per hour. In embodiments of infusion administration, infusion pump should be accurate to 5.0% of programmed rate. In embodiments of infusion administration, infusion pump should be positive pressure-driven (continuous or pulsatile). In embodiments, the reservoir or the infusion line set comprise polytetrafluoroethylene, fluorinated ethylenepropylene, polyvinylidene fluoride, polyether urethanes, polycarbonate

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urethanes, urethanes, polyurethanes, polyolefins, polyethylene, polypropylene, ethylene polymers, ethylene vinyl acetate, ethylene coacrylic acid, ethylene covinyl alcohol, polyimide, polyetheretherketone, polyaryletherketone, polysulfone, parylene, parylast, polyethylene terephthalate, polyethylene oxide, silicones, polyesters; polyolefins, polyamides, polycaprolactams, polyvinyl chloride, polyacrylates, polymethacrylates; polyureas, polyvinylhalides, polyvinylidenehalides, polyvinylethers, polyvinylaromatics, polyvinylesters, alkyd resins, polysiloxanes, epoxy resins, polyvinyl methyl ether, polyvinyl alcohol, acrylic polymers and copolymers, polyacrylonitriles, polystyrene copolymers of vinyl monomers with olefins, styrene acrylonitrile copolymers, ethylene methyl methacrylate copolymers, ethylene vinyl acetate, polyethers, rayons, cellulotics, cellulose acetate; cellulose nitrate, cellulose propionate, or any derivatives, analogs, homologues, salts, copolymers or combinations thereof. In embodiments, the reservoir or the infusion line set can be made of polyvinyl chloride, polypropylene, silicone, ethyl vinyl acetate, copolyester ether, polyolefins, or the like, or combinations thereof. In embodiments, the reservoir and infusion line set can be made of polyvinyl chloride.

In embodiments, infusion rate of the dose when administering of about 1 mcg/mL iloprost or a pharmaceutically acceptable salt thereof (including about 0.8 mcg/mL to about 1.1 mcg/mL) can be calculated as follows:

$$\text{Infusion Rate (mL/hr)} = \frac{[\text{Dose (ng/kg/min)} \times \text{Weight (kg)} \times 60 \text{ min/hr}]}{\text{Final Concentration (1,000 ng/mL)}}$$

In some embodiments, the method of the invention reduces the risk of amputation by at least 10% when compared with standard of care treatments in frostbite. In some embodiments, the method of the invention reduces the risk of amputation by at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, or at least 40% when compared with standard of care treatments in frostbite. In some embodiments, the method of the invention reduces the risk of amputation by at least 30% when compared with standard of care treatments in frostbite.

In some embodiments, standard of care for frostbite treatment includes rewarming as the first line of treatment. In some embodiments, rewarming is in conjunction with administration of aspirin and/or ibuprofen. In some embodiments, standard of care for frostbite treatment includes administration of thrombolytic therapy. In some embodiments, standard of care for frostbite treatment includes administration of tissue plasminogen activator (tPA). In some embodiments, standard of care for frostbite treatment includes administration of buflomedil.

In some embodiments, the method of the invention provides a safety profile in the subjects that is consistent with safety profile observed in systemic sclerosis subjects experiencing symptomatic digital ischemic episodes from multicenter, double-blind, randomized, placebo-controlled studies with iloprost continuous infusion. In some embodiments, the method of the invention provides a safety profile in the subjects that is consistent with safety profile observed in Example 2.

The present disclosure also relates to administering iloprost or a pharmaceutically acceptable salt thereof or a stereoisomer thereof to a frostbite patient comprising the steps of: i) transferring about 1 mL of about 100 mcg/mL concentrated solution of iloprost or pharmaceutically acceptable salt thereof into about 100 mL of 0.9% sodium chloride solution to provide an infusion solution of iloprost or pharmaceutically acceptable salt thereof; and ii) administering the infusion solution to the frostbite patient by

infusion. In embodiments, the concentrated solution of iloprost or pharmaceutically acceptable salt thereof is in a single dose or single use vial in an amount of about 1 mL. In embodiments, the single dose or the single use vial is a glass vial.

EXAMPLES

The disclosure now being generally described, it will be more readily understood by reference to the following examples which are included merely for purposes of illustration of certain aspects and embodiments of the present invention, and are not intended to limit the invention.

Example 1. Efficacy of Iloprost for Treating of Frostbite

Frostbite is an injury caused by exposure of tissue to freezing temperatures. Severe frostbite may result in amputation leading to considerable morbidity. The objective was to identify and synthesize, via systematic literature review and meta-analysis (MA), evidence published on patient-level historical amputation rates in adults with severe frostbite treated with and without iloprost.

Methods: The literature review was conducted using Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. It identified English-language articles published from Jan. 1, 2000 to Jul. 31, 2022 reporting amputation rates associated with iloprost and standard of care (SoC) treatments in severe frostbite. Study designs of interest were clinical trials (randomized and single arm), observational studies, and case series with at least 10 patients. MAs of proportions of patients with amputation were conducted. The published studies considered for the MA include:

A retrospective review of 199 patients with severe (grades 3-4) frostbite admitted to a level 1 trauma center in Colorado between 2015-2020 reported amputation rates of 49.8% in the overall population (n=199) and 59.8% in patients who did not receive thrombolytic therapy (n=127) (Carmichael et al., 2022).

A prospective observational study of 311 patients in a level 1 trauma center in Minnesota with frostbite injuries between 2014-2019 reported that, of 165 patients with severe frostbite, defined as post-rewarming perfusion deficit on imaging, 30.3% required amputation (Lacey et al., 2021). Thrombolytic therapy (tPA) was used in 75% of patients in this sample. In the subset not receiving tPA, the salvage rate was 53.4%.

A retrospective chart review study of 70 patients hospitalized for severe frostbite injuries was conducted at Chamonix Hospital in France from 1985 to 1999. In these patients, the calculated risk of amputation was reported to be

31%-67% for grade 3 frostbite (depending on the extent of the lesion to the intermediary or proximal phalanx), and 98%-100% for grade 4 frostbite (Cauchy et al., 2001).

In a retrospective review of 10 patients with deep frostbite sustained between November 2013 and March 2014 who were treated at a verified burn center in Chicago, IL, 9 (90%) patients required an amputation, which included partial or complete digital amputation or more proximal extremity amputation (Shenag et al., 2019).

Nine patients with severe frostbite from 2000 to 2017 in the University of Michigan Health System (Ann Arbor, MI) who received non-thrombolytic management were identified by retrospective review (Patel et al., 2017). In this group, all patients (100%) had at least 1 digit amputated (average digits amputated per patient was 8.6, range: 4 min-14 max).

Seventy-three frostbite patients with severe injury from 2006 to 2014 were identified in the prospectively maintained database at a single urban burn and trauma center in Minneapolis, MN (Nygaard et al., 2017). Overall, 32 patients (44%) required amputation. In the group that received SoC treatment not including thrombolytic therapy, 16/28 (50%) of patients required amputations.

Twenty-six patients with severe frostbite injury from 1995 to 2006 were treated at a single burn center in Utah (Bruen et al., 2007). Bruen et al. reported 41% of digits impacted required amputation (97 out of 243 injured digits).

In historical controls (1985-1989) at a burn center in Minnesota, Twomey et al., 2005, reported half of patients (50%) treated with standard of care required amputation (8 of 16 patients).

Determination of Risk of Amputation: Table 1 discloses the historical amputation rates in severe frostbite patients. The studies listed in Table 1 was the basis of MA for determining the historical risk of amputation. A large variability in treatments included as SoC was noted. In the US, the SoC treatment for severe frostbite is likely to be best represented by the group of studies with "Severe", "Deep" or "grade 3-4" frostbite (or an explicit definition equivalent to these), excluding studies using alprostadil. Therefore, only such studies (n=8) were included in the meta-analysis as a "base-case" scenario.

The two studies that were the biggest outliers, both with proportions of patients with amputations below the mean, were Cauchy 2000 (23%) and Rogers 2022 (29%). Cauchy 2000 was likely impacted by the unknown proportion of patients who received iloprost and/or tPA and due to the use of a different frostbite severity rating system (compared with the other studies), as well as the unknown proportion of grade 3 vs 4 frostbite. For Rogers 2022, the low rate may be in part due to the extent of thrombolytic treatment, with nearly 80% of patients receiving tPA, with the time to thrombolytic treatment in the last year of patient accrual of 6.6 hours.

TABLE 1

Historical Amputation Rates in Patients with Severe Frostbite				
Reference	Type of Study, Country, and Dates of Study	Patient Population	Treatment After Rewarming	Amputation Rate
Carceller, 2019	Retrospective observational; multinational; dates not reported	Mountaineers with frostbite (n = 91)	SoC, no tPA	68 out of 91 patients (74.7%)
Carmichael, 2022	Retrospective review; United States (Colorado); 2015-2020	Severe (grades 3-4) frostbite (n = 199)	SoC, with or without tPA (n = 199) SoC, no tPA (n = 127)	99 out of 199 patients (49.7%) 76 out of 127 patients (59.8%)

TABLE 1-continued

Historical Amputation Rates in Patients with Severe Frostbite				
Reference	Type of Study, Country, and Dates of Study	Patient Population	Treatment After Rewarming	Amputation Rate
Cauchy 2000	Retrospective review; France; 1986-1999	Severe frostbite (n = 92)	SoC	21 out of 92 patients (22.8%)
Cauchy 2011	Open label RCT; France; 1996-2008	Severe (grades 3 or 4) frostbite (n = 46)	SoC with buflomedil (n = 15)	9 out of 15 patients (60%) considered likely to receive amputation ^a
		Superficial (grade 2) frostbite (n = 1)	SoC with iloprost (n = 16)	0 patients considered likely to receive amputation ^a
			SoC with rtPA (n = 16)	3 out of 15 patients (18.7%) considered likely to receive amputation ^a
Lindford, 2017	Retrospective observational; Finland; 2013-2016	Frostbite (n = 20), of which 14 had severe frostbite	SoC (superficial) (n = 6)	0 patients with superficial frostbite underwent amputation
			SoC, no tPA or iloprost (severe) (n = 2)	Not specified
			SoC with tPA (n = 8) and with tPA and iloprost (n = 1)	4 out of 9 patients (44.4%) The patient who received both tPA and iloprost had no amputation
			SoC with iloprost (n = 3)	Not specified
			SoC with or without tPA (n = 73)	32 out of 73 patients (44%)
Nygaard, 2017	Retrospective review; United States (Minnesota); 2006-2014	Severe frostbite (n = 73)	SoC no tPA (n = 28)	16 out of 28 patients (57%)
			Thrombolytics (n = 142)	61 out of 208 patients (29.3%)
Rogers, 2022	Retrospective review; United States	Severe frostbite (n = 208)		
Shenaq, 2019	Retrospective review; United States (Illinois); 2013-2014	Cold injuries (n = 53), of which n = 10 had "deep" frostbite	SoC, no tPA	9 out of 10 patients with deep frostbite (90%) 26 out of 72 affected digits (36%) and 7 proximal amputations (trans metatarsal, Lisfranc, partial hand)

Amputation rates for patients are based on the number of patients requiring amputation of at least 1 digit. Amputation rates for digits are based on the number of digits requiring at least a partial amputation.

^a Likelihood of amputation was determined by the presence of a radiotracer anomaly in at least one digit in the bone phase of technetium scintigraphy after 8 days of treatment

SoC = standard of care;

rtPA = recombinant tissue plasminogen activator;

tPA = tissue plasminogen activator

Carceller, A. et al., (2019). Amputation Risk Factors in Severely Frostbitten Patients. International journal of environmental research and public health 16.

Carmichael, H. et al., (2022). Remote Delivery of Thrombolytics Prior to Transfer to a Regional Burn Center for Tissue Salvage in Frostbite. A Single-center Experience of 199 Patients. Journal of burn care & research: official publication of the American Burn Association 43, 54-60.

Cauchy, E. et al, (2000). The value of technetium 99 scintigraphy in the prognosis of amputation in severe frostbite injuries of the extremities. A retrospective study of 92 severe frostbite injuries. The Journal of hand surgery 25, 969-978.

Cauchy, E. et al., (2011). A controlled trial of a prostacyclin and rt-PA in the treatment of severe frostbite. N Engl J Med 364, 189-190.

Lindford, A. et al., (2017). The evolution of the Helsinki frostbite management protocol. Burns: Journal of the International Society for Burn Injuries 43, 1455-1463.

Nygaard, R. M. et al., (2017). Time Matters in Severe Frostbite. Assessment of Limb/Digit Salvage on the Individual Patient Level. Journal of burn care & research: official publication of the American Burn Association 38, 53-59.

Rogers, C. et al., (2022). The Effects of Rapid Rewarming on Tissue Salvage in Severe Frostbite Injury. Journal of burn care & research: official publication of the American Burn Association 43, 906-911.

Shenaq, D. S. et al., (2019). Urban Frostbite. Strategies for Limb Salvage. Journal of burn care & research: official publication of the American Burn Association 40, 613-619.

The literature review identified 17 primary studies; 16 were deemed eligible for the MAs of proportions. Results for pooled MA arm-level data for SoC estimated a 55% (95% confidence interval [CI]: 38%-71%; FIG. 1; RE=random effects) risk of a patient receiving at least one amputation. Tests for heterogeneity indicated high degree present with an $I^2=93.8\%$ and $p\text{-value}<0.0001$.

Efficacy-Proportion of Patients with Amputation: The systematic literature review (SLR) identified 2 studies reporting the proportion of patients with amputations after receiving iloprost (plus SoC) for frostbite (Cauchy et al., 2011; Cheguillaume, B. (2011). Controlled trial of iloprost and iloprost and rt-PA in the treatment of severe frostbite. Presented for the Award of Doctoral Degree in Medicine (Saint-Martin-d'Hères, France); Poole et al., (2021). Management of severe frostbite with iloprost, alteplase and heparin. A Yukon case series. CMAJ open 9, E585-E591). One of these studies reported amputation in zero patients out of 16 patients treated with iloprost (plus SoC), and amputation in 3 out of 16 patients who received iloprost plus rtPA (19%) (Cauchy et al., 2011; Cheguillaume, 2011), and 1 reported amputation in 4 out of 12 patients (33%) treated with iloprost (plus SoC, with or without alteplase and heparin) (Poole et al., 2021).

Sensitivity analyses reflect several ways of deriving the data (amputation rate for grade 3-4 frostbite) from the supplemental table in Cauchy 2011 (Cauchy et al., 2011). The first consideration was whether to include or exclude the patient receiving iloprost plus rtPA who was predicted to require amputation based on bone scan but was lost to follow-up. The second consideration was whether to combine the iloprost plus rtPA group with the iloprost alone group, or whether to only consider iloprost alone.

Results for pooled arm-level data for iloprost analyses found proportion of patients with amputation of 15% (95% CI: 0-45%) with high heterogeneity ($I^2=78.1\%$; $Q\text{-}p\text{-value}=0.0325$; FIG. 2; RE=random effects). All of the other scenarios provided similar results for the estimated proportion of patients with amputation (15.4% to 18.0% for different scenarios), with large error bars for this estimate and high heterogeneity (I^2 ranging from 63.1% to 78.1% for different scenarios).

Efficacy Proportion of Digits with Amputation: Data from 2 studies reporting the proportion of digits with amputations after receiving iloprost versus SoC for Grade 3/4 frostbite were included in the meta-analysis. Cauchy 2011 reported amputation in zero out of 78 digits treated with iloprost, and amputation in 40 out of 75 digits who received SoC (53.3%; Cauchy et al., 2011; Cheguillaume, 2011). When data for those receiving iloprost alone and iloprost+tPA were combined in Cauchy 2011 ($n=177$), the number of digit amputations went from zero to 3 (1.3%).

Crooks 2022 reported amputation in 62 out of 198 digits (31.1%) treated with iloprost versus 104 out of 185 digits treated with SoC (56.2%; Crooks et al., 2022). While the exact number of digits that received tPA is not specified, it was reported that only a minority of iloprost patients also received tPA (5 of 26) and a subgroup analysis showed that these patients had a higher chance of amputation compared to those receiving iloprost alone ($p<0.05$). As noted previously, there was an imbalance in the proportion of digits with Grade 3 vs Grade 4 frostbite between the 2 groups; of Grade 3-4 frostbite, Grade 4 frostbite was in only 23% of digits treated with SoC (43/185) versus 48% for iloprost (96/198). To address this imbalance, stratified odds-ratios and proportions were calculated for each study.

When the 2 studies were meta-analyzed, the odds ratio (OR) for the base-case was 0.047 (95% CI: 0.002-0.966; $p=0.047$; FIG. 3; RE=random effects). When the iloprost+tPA patients in Cauchy 2011 (Cauchy et al., 2011; Cheguillaume, 2011) were added in, the result was similar (OR: 0.034, 95% CI: 0.001, 1.12; $p=0.057$; FIG. 4; RE=random effects). For both analyses, the heterogeneity was extremely high: $I^2=82.4\%$ for the base-case and $I^2=90.3\%$ for the analysis including the patients with iloprost+tPA for both studies. This reflected that the odds ratios differed greatly between studies. While each study's individual odds ratio demonstrates a statistically significantly higher risk of digit amputation for SoC versus iloprost, the high between-study variance, and large within-study variances for Cauchy 2011 contribute to wide 95% CIs. These large variances, when combined, lead to the pooled analysis failing to show as strong a significant difference between treatments as found in the studies individually. Also, note that the employed model makes the common assumption that to the extent that underlying heterogeneity exists, it is symmetric and normally distributed. As these 95% intervals are based on such a normality assumption, they should be interpreted with caution in the presence of such heterogeneity.

The unadjusted (naïve) indirect comparisons in the meta-analysis consistently showed a 36% to 49% advantage for iloprost over SoC across a variety of scenarios, with the base-case result (targeting studies for which there was confidence that all patients had severe frostbite and did not receive alprostadil) with an advantage of 39% [95% CI: 6-73%, $p=0.02$]. That is, comparisons between iloprost and SoC showed a 39% reduction in amputation for iloprost.

Three iloprost studies reported safety data. Safety events associated with iloprost were generally mild and included headache, hot flushes/flushing, palpitations/tachycardia, and nausea and vomiting. These were managed symptomatically or by lowering iloprost dose, and rarely resulted in iloprost discontinuation.

Conclusion: Our literature review of the treatment of severe frostbite with either iloprost or SoC found a substantial rate of amputation with SoC and, comparatively, a significant reduction in amputation rate in patients treated with iloprost.

Example 2. A Multicenter, Double-Blind, Randomized, Placebo-Controlled, Study Evaluating Intravenous Iloprost in Subjects with Symptomatic Raynaud's Phenomenon Secondary to Systemic Sclerosis (SSc)—Evaluation of Safety of Iloprost

This was a multicenter, double-blind, randomized, placebo-controlled study to provide an initial evaluation of the effect of iloprost on the symptomatic relief of RP attacks in subjects with SSc.

Subjects were randomized in a 1:1 ratio to iloprost injection for IV use or placebo. Randomization was stratified based on the use of phosphodiesterase inhibitors at screening. Study drug administration began on Day 1, and subjects received study drug for 5 consecutive days (e.g., Monday through Friday) as an IV infusion over 6 hours each day via a peripheral line (NovaCath™ Integrated IV Catheter System) or a peripherally inserted central catheter using an infusion pump.

Subjects must have had a systolic blood pressure ≥ 85 mmHg (sitting position) prior to study drug administration each day of administration. On Day 1, study drug was initiated at a starting dose of 0.5 ng/kg/min, and dose increases occurred every 30 minutes (± 5 minutes) in incre-

ments of 0.5 ng/kg/min up to 2.0 ng/kg/min or the individual tolerated dose. If dose-limiting adverse events (e.g., headache, flushing, jaw pain, myalgia, nausea, or vomiting) occurred that could not be tolerated by the subject, or if the subject experienced symptomatic hypotension, then the dose was reduced in a step-wise manner by 0.5 ng/kg/min every 30 minutes (+5 minutes) until a tolerated dose was determined or the infusion was stopped until the symptoms resolved at which point the study drug was reinitiated at a previously tolerated dose. If symptomatic hypotension or a dose-limiting adverse event occurred during administration of iloprost at the starting dose (i.e., 0.5 ng/kg/min), the infusion was reduced to 0.25 ng/kg/min. If the dose of 0.25 ng/kg/min was not tolerated due to symptomatic hypotension or if a dose-limiting adverse event occurred, the study drug was discontinued, and re-initiation of the infusion could have been attempted after the event had resolved or had been treated. Blood pressure and heart rate were monitored 15 minutes (± 5 minutes) prior to and after all dose changes. The maximum tolerated dose was maintained for the remaining 6-hour daily period. At the end of the 6-hour infusion period, the dose was stopped. Subjects were to be monitored for up to 1 hour after completion of study drug infusion (i.e., vital signs were obtained 15 minutes [± 5 minutes] and 1 hour [± 15 minutes] after completion of the infusion).

On Days 2 to 5, the infusion was started using the highest infusion rate tolerated on the previous day without up- or down-titration, unless the subject did not tolerate the infusion or adverse events occurred that could not be tolerated by the subject and necessitated a reduction in the dose. Vital signs were measured prior to study drug administration and at 15 minutes (± 5 minutes) prior to and after all dose changes during the infusion. Additionally, vital signs were monitored at 15 minutes (± 5 minutes) and 1 hour (± 15 minutes) after completion of the 6-hour infusion.

During the treatment period (Days 1 to 5), while subjects received study drug, the electronic patient-reported outcomes (ePRO) diary was not completed. No study assessments were performed on the 2 days following the end of treatment (Days 6 and 7 [i.e., Saturday and Sunday]) to allow the subject to rest and return to a schedule of normal daily living activity following the 5 days of infusions.

Subjects were contacted via telephone on Day 8 to ensure they resumed completion of the daily ePRO diary; subjects completed the ePRO diary from Day 8 through Day 21. On Day 22, subjects returned to the clinic for post-treatment evaluations. A follow-up visit occurred 30 days after the last administration of study drug (Day 35).

Subjects who discontinued study drug early remained in the study (unless the subject withdrew consent) and completed the daily ePRO diary from Day 8 to Day 21, including clinical laboratory assessments on the remaining missed infusion days as well as post-treatment study assessments.

The total duration of the study for a subject was up to approximately 9 weeks.

Number of subjects: Randomized—34 subjects; Completed—34 subjects

Diagnosis and Main Criteria for Inclusion: The population for this study included male and female subjects ≥ 18 years of age who met the following criteria: had a diagnosis of SSc, as defined by the 2013 American College of Rheumatology criteria/European League Against Rheumatism criteria; had a diagnosis or history of RP, self-reported or reported by a physician, with at least a 2-phase color change in figure(s) of pallor, cyanosis, and/or reactive hyperemia in response to cold exposure or emotion; and had a minimum

of 10 symptomatic RP attacks, documented in the ePRO diary, that occurred over at least 3 separate days of the 5-day eligibility period.

Investigational Product and Comparator Information: Iloprost injection for IV use and matching placebo were supplied in vials packaged in a blinded study drug kit (10 vials per kit). The iloprost and placebo vials were identical, except 100 mcg of iloprost was added to the active study drug vials. The drug product was diluted with sodium chloride 0.9% in a drug reservoir (IV bag) prior to use.

Safety parameters included adverse events, physical examination findings, vital sign measurements (heart rate and blood pressure), 12-lead electrocardiogram (ECG) findings, and standard clinical laboratory measurements (chemistry and hematology).

Summary of Safety Results:

No subjects experienced a serious adverse event (SAE), adverse event of special interest (AESI), treatment-emergent adverse event (TEAE) leading to discontinuation of study drug, or TEAE leading to death in this study.

Overall, 31 (91.2%) subjects experienced a TEAE: 14 (82.4%) subjects in the placebo group and 17 (100.0%) subjects in the iloprost group. The majority of TEAEs were considered mild to moderate in severity. Two (11.8%) subjects in the iloprost group experienced TEAEs that were considered severe by the Investigator.

The most commonly reported TEAEs (based on the total number of subjects who experienced this TEAE) were headache (21 [61.8%] subjects total: 5 [29.4%] subjects in the placebo group and 16 [94.1%] subjects in the iloprost group), nausea (14 [41.2%] subjects total: 3 [17.6%] subjects in the placebo group and 11 [64.7%] subjects in the iloprost group), abdominal pain (5 [14.7%] subjects total: 1 [5.9%] subject in the placebo group and 4 [23.5%] subjects in the iloprost group), and flushing (5 [14.7%] subjects total: 0 [0.0%] subjects in the placebo group and 5 [29.4%] subjects in the iloprost group).

There were no clinically significant changes in chemistry or hematology laboratory values during the study.

There were no TEAEs related to vital signs, physical examination findings, or ECGs in this study.

There were no hypertensive or hypotensive events reported in this study. There was no bradycardia or tachycardia reported in this study.

Conclusions:

The results confirmed that the iloprost dosing titration algorithm could be safely used in subjects. All adverse events related to the study drug as assessed by the Investigator were expected and consistent with the known safety profile of iloprost. There were no deaths, SAEs, AESIs, or TEAEs leading to study drug discontinuation during this study.

Example 3. Solution Stability of Fully Diluted Ready-to-Use Iloprost

Iloprost 100 $\mu\text{g/mL}$ injection product is administered to patients by IV bag infusion over a period of six hours. The fully diluted ready to use iloprost is prepared in pharmacy cleanroom using sterile preparation standards (USP Chapter <797>) by pharmacy trained and licensed staff. The fully diluted ready to use drug product may be prepared several hours ahead of treatment with storage at ambient conditions or several days ahead of treatment with storage at refrigerated ($2-8^{\circ}\text{C}$.) conditions. The IV bags are prepared using empty sterile IV bags which are filled with 99 mL of 0.9% NaCl and 1.0 mL of iloprost 100 $\mu\text{g/mL}$ injection product.

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The bags are then manually mixed gently to provide a uniform solution. Administration IV bags were prepared and stored at several conditions with routine sampling for assay testing for stability under i) 25° C./60% RH, ii) 5° C. for 5 days to 25° C./60% RH, iii) 5° C. for 8 days to 25° C./60% RH.

IV Bag Sample Preparation: Two 50 mL syringes with 18-gauge needles were assembled. 1.0 mL syringe with 18-gauge needle was assembled. Using a 50 mL syringe, 50 mL of 0.9% NaCl was transferred into an empty IV bag (sterile, ICU Medical) using the injection port. Using a second 50 mL syringe, 49 mL of 0.9% NaCl was transferred into the same IV bag using the injection port resulting in 99 mL of solution. Using the 1 mL syringe, 1.0 mL of iloprost 100 µg/mL injection product was transferred into the IV bag using the injection port. The needle was removed and the port was securely closed. The IV bag was gently inverted

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have passed for condition iii), the IV bags were removed from the chamber and allowed to equilibrate to room temperature. Each IV bag was mixed by inversion several times. Sampling from each IV bag was taken for HPLC analysis. Then the IV bags were placed in the 25° C./60% RH stability chamber and sampled at certain time points as discussed herein.

Sampling time for condition ia): 2 h, 4 h, 6 h, 8 h, and 24 h

Sampling time for ib): 8 h, 12 h, 48 h, and 96 h

Sampling time for condition ii): 5 days (at the end of storage in 2-8° C. stability chamber), 5 days and 8 h, and 5 days and 12 h

Sampling time for condition iii): 8 days (at the end of storage in 2-8° C. stability chamber), 8 days and 8 h, and 8 days and 12 h

Sample Analysis: Samples were analyzed by HPLC using a validated method (validation protocol not shown).

HPLC Column:	Waters Spherisorb ODS-2, 125 mm, ×4.6 mm, 3 µm, P/N PSS832116		
Column Temperature:	20° C.		
Sample Temperature:	5° C.		
Flow Rate:	1.0 mL/min		
Injection Volume:	100 µL		
UV (DAD or MWD):	200 nm, bandwidth 4 nm; Identification by UV Spectrum 200-400 nm		
UV (VWD):	200 nm		
Run Time:	55 Minutes		
Needle Rinse:	(75:25 Milli-Q™ Water:ACN)		
Mobile Phase A:	(8 g/L BCD in 33% ACN, pH 2.0)		
Mobile Phase B:	(50:50:0.1 Milli-Q™ Water: ACN:H3PO4)		

	Time (minutes)	% Mobile Phase A	% Mobile Phase B
Gradient:	0	100	0
	30	100	0
	35	5	95
	50	5	95
	50.1	100	0
	55	100	0

several times to mix. During inversion, it was occasionally paused to squeeze the injection port area to ensure liquid rinsed into area several times.

Sampling Procedure: For sampling, a 3 mL syringe was assembled with an 18-gauge needle. Using a 3 mL syringe, 2 mL of IV bag solution was withdrawn through the injection port and 1 mL was transferred to a HPLC vial for neat injection and the other 1 mL was transferred to a back-up HPLC vial. HPLC vials were stored at refrigerated conditions 2-8° C.

Sample Storage and Sampling: For each testing conditions (i-iv discussed above), 12 IV bag samples were prepared (6 IV bags for condition ia)—see below for sampling time). For condition i), all 12 IV bags were then placed into 25° C./60% RH stability chamber ensuring each bag is resting on its largest surface area side. After the first time point, the IV bags were removed from the chamber and each IV bag was mixed by inversion several times. Sampling from each IV bag was taken for HPLC analysis. After sampling, the IV bags were placed back into 25° C./60% RH stability chamber ensuring each bag is resting on its largest surface area side, until next sampling time. At each sampling time, the IV bags were removed from the chamber and each IV bag was mixed by inversion several times before samples were removed.

For conditions ii) and iii), all 12 IV bags for each testing condition were placed into a 2-8° C. stability chamber ensuring each bag is resting on its largest surface area side. After 5 days have passed for condition ii) or after 8 days

The concentration of each sample per the equation below:

Sample Concentration (µg/mL) =

$$\frac{Rsam}{Rstd} \times Cstd \text{ (mg/mL)} \times \frac{Vpip \text{ (mL)}}{Vstd \text{ (mL)}} \times 1000 \text{ µg/mg}$$

Rstd=Average sum of the areas of the Iloprost peaks from five Standard Solution injections made during system suitability.

Rsam=Sum of the areas of the Iloprost peaks from the sample solution injection

Cstd (mg/mL)=Concentration of Iloprost Stock Standard Solution in mg/mL

Vstd (mL)=Volume of Standard Solution in mL (10 mL)

Vpip (mL)=Volume of Iloprost Stock Standard Solution pipetted into the Standard Solution in mL (0.1 mL)

The % Initial (initial concentration) of each stored sample was determined per the equation below:

$$\% \text{ Initial} = \frac{\text{Stored Sample Concentration (µg/mL)}}{\text{Initial (T = 0) SampleConcentration (µg/mL)}} \times 100$$

The results from the IV Bag stability study under condition ia (25° C./60% RH stability; sample time 2 h, 4 h, 6 h, 8 h, and 24 h) are presented in Table 1A. The % initial results for all of the IV bag preparations exhibited a range of

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94.2-94.9%. These results were as expected from the nature of the IV bag preparation which was designed to mimic the clinical procedure. The IV bags were adjusted for volume to a target weight and then spiked with drug product using a disposable sterile 1 mL syringe equipped with a disposable sterile 18 gauge needle. A combination of variability in the weight of the IV bags and variability in the volume accuracy of the syringes and needles was expected to affect the accuracy of the spiking procedure. Each sample met the specification criterion of 90.0-110.0% and each stored sample was also within 90.0-110.0% of the initial result for the preparation.

TABLE 1A

24 hour stability at 25° C./60% RH			
Sample No.	Time-point (h)	Iloprost (%)	% Initial
1	Initial	94.9	
	2	95.3	100.4
	4	96.7	101.9
	6	95.4	100.6
	8	94.8	99.9
	24	97.0	102.2
2	Initial	94.2	
	2	95.0	100.8
	4	95.2	101.0
	6	95.7	101.6
	8	94.3	100.0
	24	95.5	101.4
3	Initial	94.8	
	2	94.2	99.4
	4	94.3	99.5
	6	95.0	100.2
	8	94.8	100.0
	24	96.7	102.0
4	Initial	94.6	
	2	94.8	100.2
	4	94.3	99.6
	6	95.1	100.5
	8	94.6	100.0
	24	94.7	100.1
5	Initial	94.4	
	2	94.2	99.7
	4	94.7	100.3
	6	93.7	99.3
	8	96.2	101.9
	24	93.4	98.9
6	Initial	94.3	
	2	94.7	100.5
	4	95.5	101.3
	6	95.9	101.7
	8	95.3	101.1
	24	94.8	100.5

The results from the IV Bag stability study under condition ib (25° C./60% RH stability; sample time 8 h, 12 h, 48 h, and 96 h) are presented in Tables 1B and 1C. Each sample met the specification criterion of 90.0-110.0% and each stored sample was also within 90.0-110.0% of the initial result for the preparation.

TABLE 1B

96 hour stability at 25° C./60% RH (Batch 1)			
Time Point	Preparation	Iloprost (ng/mL)	% Initial
Initial (T = 0)	1	977.49068	—
	2	968.66746	
	3	942.80790	
	4	994.14641	
	5	981.91733	
	6	975.78016	

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TABLE 1B-continued

96 hour stability at 25° C./60% RH (Batch 1)			
Time Point	Preparation	Iloprost (ng/mL)	% Initial
8 hr	1	973.57406	100
	2	1009.87852	104
	3	979.46989	104
	4	1002.63999	101
	5	977.22453	100
	6	1007.06584	103
12 hr	1	1004.19286	103
	2	1008.58278	104
	3	978.15686	104
	4	1042.47061	105
	5	975.06820	99
	6	1005.44196	103
48 hr	1	1003.58363	103
	2	995.23565	103
	3	968.03688	103
	4	1001.02390	101
	5	984.61614	100
	6	977.30241	100
96 hr	1	1013.28284	104
	2	1017.49428	105
	3	970.92026	103
	4	1014.14964	102
	5	994.07339	101
	6	971.53175	100

TABLE 1C

96 hour stability at 25° C./60% RH (Batch 2)			
Time Point	Preparation	Iloprost (ng/mL)	% Initial
Initial (T = 0)	1	980.86863	—
	2	990.69772	
	3	980.65678	
	4	994.36439	
	5	1000.53816	
	6	1020.67367	
8 hr	1	995.76595	102
	2	1002.10267	101
	3	994.55360	101
	4	1003.76351	101
	5	998.77363	100
	6	983.28166	96
12 hr	1	993.30159	101
	2	984.11952	99
	3	984.12486	100
	4	989.62285	100
	5	994.77764	99
	6	1004.99733	98
48 hr	1	993.66797	101
	2	984.23731	99
	3	994.36502	101
	4	1031.78023	104
	5	1005.03600	100
	6	1014.71909	99
96 hr	1	1002.16769	102
	2	979.52872	99
	3	988.97656	101
	4	1004.97292	101
	5	996.34817	100
	6	1027.43687	101

The results from the IV Bag stability study under condition ii (5° C. for 5 days to 25° C./60% RH) are presented in Tables 1D and 1E. Each sample met the specification criterion of 90.0-110.0% and each stored sample was also within 90.0-110.0% of the initial result for the preparation.

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TABLE 1D

5° C. for 5 days to 25° C./60% RH (Batch 1)			
Time Point	Preparation	Iloprost (ng/mL)	% Initial
Initial (T = 0)	1	1007.32936	—
	2	953.59154	
	3	1014.73720	
	4	1019.54404	
	5	1036.25326	
5° C. for 5 days	6	996.56811	
	1	1049.58766	104
	2	931.91928	98
	3	1023.92132	101
	4	1020.05717	100
5° C. for 5 days	5	1024.17465	99
	6	1000.90608	100
5° C. for 5 days	1	1002.17085	99
	2	957.27263	100
	3	993.78991	98
	4	1040.15942	102
	5	1025.38946	99
25° C./60% RH for 8 hr	6	1010.55688	101
	1	976.52119	97
	2	943.23563	99
	3	1015.06188	100
	4	1030.59268	101
5° C. for 5 days	5	1010.05028	97
	6	982.68256	99

TABLE 1E

5° C. for 5 days to 25° C./60% RH (Batch 2)			
Time Point	Preparation	Iloprost (ng/mL)	% Initial
Initial (T = 0)	1	1011.00891	—
	2	986.02971	
	3	1021.90645	
	4	1021.61212	
	5	1001.14785	
5° C. for 5 days	6	988.43978	
	1	1026.46301	102
	2	1008.56452	102
	3	1024.57969	100
	4	1029.90237	101
5° C. for 5 days	5	1011.12281	101
	6	987.37008	100
5° C. for 5 days	1	1023.49747	101
	2	973.07617	99
	3	1013.01926	99
	4	1011.71269	99
	5	996.65926	100
25° C./60% RH for 8 hr	6	970.82806	98
	1	1022.38730	101
	2	963.28102	98
	3	1001.36417	98
	4	1003.23343	98
5° C. for 5 days	5	1016.70001	102
	6	985.93577	100

The results from the IV Bag stability study under condition iii (5° C. for 8 days to 25° C./60% RH) are presented in Tables 1F and 1G. Each sample met the specification criterion of 90.0-110.0% and each stored sample was also within 90.0-110.0% of the initial result for the preparation.

TABLE 1F

5° C. for 8 days to 25° C./60% RH (Batch 1)			
Time Point	Preparation	Iloprost (ng/mL)	% Initial
Initial (T = 0)	1	970.11915	—
	2	966.13069	
	3	989.59242	

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TABLE 1F-continued

5° C. for 8 days to 25° C./60% RH (Batch 1)			
Time Point	Preparation	Iloprost (ng/mL)	% Initial
5° C. for 5 days	4	959.55241	
	5	964.81219	
	6	957.61598	
	1	994.43539	103
	2	953.18657	99
5° C. for 5 days	3	978.41500	99
	4	953.14379	99
	5	984.18557	102
	6	956.73023	100
	1	963.52003	99
5° C. for 8 days	2	955.94360	99
	3	967.83409	98
	4	938.70012	98
	5	1032.99481	107
	6	961.84780	100
25° C./60% RH for 8 hr	1	980.47291	101
	2	964.06225	100
	3	986.44359	100
	4	967.53103	101
	5	964.22470	100
5° C. for 8 days	6	952.01922	99

TABLE 1G

5° C. for 8 days to 25° C./60% RH (Batch 2)			
Time Point	Preparation	Iloprost (ng/mL)	% Initial
Initial (T = 0)	1	979.57170	—
	2	983.59476	
	3	998.93472	
	4	1008.63220	
	5	1025.59384	
5° C. for 5 days	6	992.74821	
	1	970.13301	99
	2	957.91301	97
	3	1019.54124	102
	4	1006.94224	100
5° C. for 5 days	5	993.60327	97
	6	1014.48560	102
5° C. for 8 days	1	983.93098	100
	2	964.53372	98
	3	1004.08143	101
	4	997.60330	99
	5	1010.96396	99
25° C./60% RH for 8 hr	6	1010.24659	102
	1	967.45808	99
	2	964.40986	98
	3	1006.95543	101
	4	1011.04730	100
5° C. for 8 days	5	1012.62453	99
	6	1008.92483	102

During the study, three unknown peaks in chromatograms were observed; however, all of these unknown peaks were present in control or stressed blank IV bag samples indicating that all unknown peaks were IV bag or saline related and were not caused by Iloprost drug product. The unknown peaks did not interfere with quantitation of Iloprost peaks and therefore the administration stability study is not considered to be impacted by the presence of the peaks.

These results indicate that the iloprost drug product exhibits suitable stability in the IV bags to support IV bag treatment and administration design.

Example 4. Study of Microbial Attribute of Iloprost 100 µg/mL Injection Composition

The aim of this in-use stability study is to provide data reflecting the microbiological quality of iloprost 100 µg/mL

injection composition and iloprost placebo injection composition after their preparation and storage under controlled storage conditions for a specific period of time. These composition preparation and storage simulates their preparation and storage conditions in compounding pharmacy prior to patient administration.

Iloprost 100 µg/mL injection composition and iloprost placebo injection composition samples are prepared (diluted in saline IV bags), stored under controlled conditions and tested at specific time points (day 0, day 10, and day 16). The prepared samples are assessed side-by-side with prepared samples that have been inoculated with a low level of bioburden (10-100 CFU/mL). The low level of bioburden is intended to simulate a microbial contamination at the time of container closure penetration. This microbial challenge study helps determine if the diluted ready to use products have any growth-promoting properties. Growth-promoting properties of these two drug products are assessed according to the USP <51> guidance.

Microbial counts of the inoculated products at time zero is compared to their microbial counts at day 10 and day 16. If the microbial counts on day 10 and day 16 are not higher than 0.5 log₁₀ unit compared to the starting microbial counts then, the drug products are considered non-growth-promoting and the storage conditions (time and temperature) are deemed appropriate for ensuring that the diluted ready to use drug products are safe.

This in-use stability study is designed and executed per the following guidelines:

The current United States Pharmacopeia General Chapters USP <51>, "Antimicrobial Effectiveness testing"

The CDER guidance on the subject by Metcalfe John W. (2009)

The CDER Microbiology Issues: A deep Dive by Candace Gomez-Broughton, "Aseptic Processing of Biological Products: Current Regulatory Issues", August 2018

The World Health Organization guidance on the subject: WHO Technical Report Series, No. 863, 1996, Annex 5 The Procedures Outlined in this Protocol

Scope: This protocol applies to microbiological testing in support of the microbiological stability of the diluted ready to use iloprost 100 µg/mL injection composition and iloprost placebo injection composition. The stability of these two drug products is assessed after penetration of the container and closure system for dose preparation and storage under controlled conditions. The preparation and storage conditions of the two drug products simulate the preparation and storage prior to patient administration. This protocol describes the study design and Bioburden testing using representative samples of iloprost 100 µg/mL injection composition and iloprost placebo injection composition samples.

Interpretation of Results: Initial microbial count (CFU/mL) for each inoculated sample is defined as the microbial count of that sample at time zero and determined by filtration test method. Growth or its absence thereof for a sample/microorganism combination at a specific time point is assessed in relation to time zero. Microbial count is determined for each microorganism at time zero and the log₁₀ of that microbial count will be calculated. Microbial count for each test sample will be determined at the specific time point, log₁₀ of that microbial count is calculated and compared to its microbial count log₁₀ value at time zero. If the difference between the two log₁₀ values is not more than half log₁₀ unit, the sample is not displaying microbial growth increase.

Iloprost 100 µg/mL injection composition preparation: Two 50 mL syringes with 18-gauge needles are assembled.

1.0 mL syringe with 18-gauge needle is assembled. Using a 50 mL syringe, 49 mL of 0.9% NaCl is transferred into an empty IV bag using the injection port. Using a second 50 mL syringe, 50 mL of 0.9% NaCl is transferred into the same IV bag using the injection port. Using the 1 mL syringe, 1.0 mL of iloprost 100 µg/mL injection product is transferred into the IV bag using the injection port. The needle was removed and the port was securely closed. The IV bag is gently inverted several times to mix. During inversion, occasionally paused to squeeze the injection port area to ensure liquid rinsed into area several times. Prepare total of 7 IV bags by this method.

Microbial Inoculation:

S. aureus, *B. subtilis*, *P. aeruginosa*, *C. albicans*, *A. brasiliensis*, and *E. coli* are used. Patheon frozen cultures, freshly harvested suspensions, or enumerated lyophilized microorganism preparations are used. Microorganisms that are not more than five passages removed from the master seed lot are used.

Test Samples: Using the appropriate syringe size and an 18-gauge needle, inoculate (inject through injection port) each IV bag and each IV bag of positive control with the appropriate microorganism suspension so that the inoculated IV bag contains between 5-10 CFU/mL (for a total of 500 to 1000 CFU/IV bag). Gently invert each IV bag several times to mix. During inversion, pause occasionally to squeeze the injection port area to ensure liquid rinses into area several times.

Inoculum Verification (perform in duplicate for each microorganism at time zero): Add 1 mL from each positive control sample to each of two 100×15 mm Petri dishes. Pour approximately 25 mL Lethen agar cooled to ≤45° C. into each of two Petri dishes and swirl to mix. Allow the agar to solidify and invert the plates to incubate. Incubate the bacteria plates at 30 to 35° C. for 3-5 days. Incubate *Candida albicans* plates at 20-25° C. for 3 to 5 days and incubate the *Aspergillus brasiliensis* plates at 20-25° C. for 3 to 7 days.

Sampling: Day 0 samples are taken immediately after the IV bags were prepared. Test samples are taken after 10 days and 16 days storage of the inoculated bags at 2-8° C. At the specified time point, assemble a 50 mL syringe with 18-gauge needle. Using the 50 mL syringe, withdraw 25 mL from each of the inoculated IV bags through the injection port and transfer to an appropriate size sterile test tube with cap. After taking the test samples, place the inoculated IV bags at 2-8° C. until the next sampling time point.

Testing Samples: Test the inoculated product samples, positive controls, product negative controls (iloprost and diluent), and diluent negative control. Each sample is tested in duplicate (two filters will be used). Prewash each of two sterile filter membranes with 100 mL Dilution Fluid D. Add a 10 mL aliquot of the test sample (inoculated drug product, product negative control, positive control or diluent negative control) to each filter unit and filter through. Wash each filter with 3×100 mL aliquots of Dilution Fluid D. Aseptically transfer the two filters to two separate Lethen Agar plates. Allow the agar to solidify and incubate as follows:

- The inoculated product samples, and the positive control samples containing bacteria at 30 to 35° C. for 3 to 5 days.
- The inoculated product samples, and the positive control samples containing *Candida albicans* at 20-25° C. for 3 to 5 days
- The inoculated product samples, and the positive control samples containing *Aspergillus brasiliensis* at 20-25° C. for 3 to 7 days.

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d. The product and diluent negative controls: From each set, incubate one plate at 30-35° C. and one plate at 20-25° C. for as long as the test samples but not more than 7 days.

Rinse Fluid Negative Controls: Add 100 mL Dilution Fluid D to each of 2 sterile filter units and filter through. Aseptically transfer the two filters to two plates of Lethen agar. Incubate one plate at 30-35° C. and the other plate at 20-25° C. for as long as the test samples but not more than 7 days.

Agar Negative Controls: Incubate one plate of Lethen agar at 30-35° C. and the other plate at 20-25° C. for as long as the test samples but not more than 7 days.

Analysis: Count the Colony Forming Units (CFU) on each plate per SOP-QC-280 and calculate the arithmetic mean for each set of two plates. Calculate the percent recovery by dividing the mean of the product plate counts (CFU) by the mean of the positive control plate counts (CFU). Multiply by 100 and round to the nearest whole number.

$$\% \text{ Recovery} = \frac{\text{Mean of inoculated Product plates count (CFU)} \times 100}{\text{Mean of inoculated positive controls plates count (CFU)}}$$

Acceptance Criteria:

The percent recovery of the product test plates, at time zero, should be at least 50% of the positive controls mean. If lower recoveries (less than 50%) are observed, modify the method in order to overcome inhibition by utilizing one or more of the following:

Incorporate Neutralizing/Dispersing Agents into the Plating Medium or Rinse Fluid
Utilize Different Media

Each inoculum verification plate must have ≥ 1 CFU and ≤ 100 CFU

There must be no growth on the agar and rinse fluid negative controls. A media or rinse fluid negative control failure requires an investigation to evaluate the impact to the testing described herein.

Results:

Microbial Enumeration Test Method Suitability. Suitability of the microbial enumeration test method was assessed at time zero by comparing the microbial recoveries from the diluted-ready-to use products to the microbial recoveries from the diluent (positive control). The method was considered suitable if the following criteria were met:

1. Each inoculum verification plate must have ≥ 1 CFU and ≤ 100 CFU
2. There must be no growth on the product negative controls, agar negative controls and rinse fluid negative controls.
3. The microbial recovery from the diluted-ready-to use products is at least 50% of the recovery from the diluent (positive control)

Each of the inoculum verification plate (*S. aureus*, *B. subtilis*, *P. aeruginosa*, *C. albicans*, *A. brasiliensis*, and *E. coli*) counts met the acceptance criteria: ≥ 1 and ≤ 100 CFU. All media, diluent and dilution fluid D negative controls exhibited no growth. All product negative controls exhibited no growth.

The recoveries of *S. aureus*, *B. subtilis*, *P. aeruginosa*, *C. albicans*, *A. brasiliensis*, and *E. coli* from iloprost 100 µg/mL injection composition and iloprost placebo injection composition samples, respectively, were all above 50% of the positive control recovery.

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Assessment of growth-promoting properties of iloprost 100 µg/mL injection composition and iloprost placebo injection composition samples. Microbial counts were determined for each sample/microorganism combination at each time point, and the log 10 of the microbial count mean was calculated. The difference between the log 10 value (time X and time zero) was used to assess if the product is growth-promoting or not. The products were deemed non-growth promoting as assessed by this testing if the following criteria were met:

1. In accordance with USP <51>, for each tested microorganism there must be no microbial count increase that is higher than 0.5 log 10 relative to microbial counts at time zero.
2. There must be no growth on the product negative controls, agar negative controls and rinse fluid negative controls.

Results for each microorganism are summarized in the Tables 2A-2F.

$$\text{Log}_{10} \text{ difference} = \log_{10} \text{ value at time X} - \log_{10} \text{ value at time 0}$$

For each of *S. aureus*, *B. subtilis*, *P. aeruginosa*, *C. albicans*, *A. brasiliensis*, and *E. coli*, the log₁₀ difference for each time point was not greater than 0.5 log 10 unit from time zero. Ready-to use iloprost 100 µg/mL injection composition and iloprost placebo injection composition do not promote *S. aureus*, *B. subtilis*, *P. aeruginosa*, *C. albicans*, *A. brasiliensis*, and *E. coli* growth when stored at 2-8° C. for up to 16 days.

The results presented in Tables 2A-2F show that the tested ready-to use iloprost 100 µg/mL injection composition and iloprost placebo injection composition do not display any growth-promoting properties towards the tested microorganisms: *S. aureus*, *B. subtilis*, *P. aeruginosa*, *C. albicans*, *A. brasiliensis*, and *E. coli*. This test meets the acceptance criteria for the assessment of growth-promoting properties of both iloprost 100 µg/mL injection composition and iloprost placebo injection composition.

TABLE 2A

Log ₁₀ comparison for <i>A. brasiliensis</i>				
Batch	Time point	Mean Microbial Count (CFU)	Log ₁₀ of Microbial Count	Log difference compared to Time 0
1	Day 0	50	1.6990	
	Day 10	45	1.6532	-0.0458
	Day 16	47	1.6721	-0.0269
2	Day 0	47	1.6721	
	Day 10	50	1.6990	0.0269
	Day 16	47	1.6721	0
3	Day 0	44	1.6435	
	Day 10	57	1.7559	0.1124
	Day 16	48	1.6812	0.0377
4	Day 0	48	1.6812	
	Day 10	45	1.6532	-0.0280
	Day 16	59	1.7709	0.0897

TABLE 2B

Log ₁₀ comparison for <i>C. albicans</i>				
Batch	Time point	Mean Microbial Count (CFU)	Log ₁₀ of Microbial Count	Log difference compared to Time 0
1	Day 0	30	1.4771	
	Day 10	5	0.6990	-0.7781
	Day 16	1	0	-1.4771

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TABLE 2B-continued

Log ₁₀ comparison for <i>C. albicans</i>				
Batch	Time point	Mean Microbial Count (CFU)	Log ₁₀ of Microbial Count	Log difference compared to Time 0
2	Day 0	48	1.6812	
	Day 10	2	0.3010	-1.3802
	Day 16	1	0	-1.6812
3	Day 0	44	1.6435	
	Day 10	2	0.3010	-1.3425
	Day 16	0	No growth	N/A
4	Day 0	38	1.5798	
	Day 10	1	0	-1.5798
	Day 16	1	0	-1.5798

TABLE 2C

Log ₁₀ comparison for <i>E. coli</i>				
Batch	Time point	Mean Microbial Count (CFU)	Log ₁₀ of Microbial Count	Log difference compared to Time 0
1	Day 0	61	1.7853	
	Day 10	26	1.4150	-0.3703
	Day 16	16	1.2041	-0.5812
2	Day 0	58	1.7634	
	Day 10	30	1.4771	-0.2863
	Day 16	12	1.0792	-0.6842
3	Day 0	32	1.5051	
	Day 10	12	1.0792	-0.4259
	Day 16	9	0.9542	-0.5509
4	Day 0	73	1.8633	
	Day 10	22	1.3424	-0.5209
	Day 16	11	1.0414	-0.8219

TABLE 2D

Log ₁₀ comparison for <i>P. aeruginosa</i>				
Batch	Time point	Mean Microbial Count (CFU)	Log ₁₀ of Microbial Count	Log difference compared to Time 0
1	Day 0	31	1.4914	
	Day 10	0	No growth	N/A
	Day 16	0	No growth	N/A
2	Day 0	26	1.4150	
	Day 10	1	0	-1.4150
	Day 16	0	No growth	N/A
3	Day 0	25	1.3979	
	Day 10	1	0	-1.3979
	Day 16	0	No growth	N/A
4	Day 0	33	1.5185	
	Day 10	0	No growth	N/A
	Day 16	0	No growth	N/A

TABLE 2E

Log ₁₀ comparison for <i>S. aureus</i>				
Batch	Time point	Mean Microbial Count (CFU)	Log ₁₀ of Microbial Count	Log difference compared to Time 0
1	Day 0	48	1.6812	
	Day 10	16	1.2041	-0.4771
	Day 16	8	0.9031	-0.7781
2	Day 0	39	1.5911	
	Day 10	18	1.2553	-0.3358
	Day 16	3	0.4771	-1.114
3	Day 0	52	1.7160	
	Day 10	15	1.1761	-0.5399
	Day 16	6	0.7782	-0.9378

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TABLE 2E-continued

Log ₁₀ comparison for <i>S. aureus</i>				
Batch	Time point	Mean Microbial Count (CFU)	Log ₁₀ of Microbial Count	Log difference compared to Time 0
4	Day 0	51	1.7076	
	Day 10	20	1.3010	-0.4066
	Day 16	2	0.3010	-1.4066

TABLE 2F

Log ₁₀ comparison for <i>B. subtilis</i>				
Batch	Time point	Mean Microbial Count (CFU)	Log ₁₀ of Microbial Count	Log difference compared to Time 0
1	Day 0	67	1.8261	
	Day 10	32	1.5051	-0.3210
	Day 16	15	1.1761	-0.6500
2	Day 0	68	1.8325	
	Day 10	30	1.4771	-0.3554
	Day 16	14	1.1461	-0.6864
3	Day 0	53	1.7243	
	Day 10	23	1.3617	-0.3626
	Day 16	15	1.1761	-0.5482
4	Day 0	55	1.7404	
	Day 10	26	1.4150	-0.3254
	Day 16	11	1.0414	-0.6990

The disclosures of all publications, patents, patent applications and published patent applications referred to herein by an identifying citation are hereby incorporated herein by reference in their entirety.

In the case of any conflict between a cited reference and this specification, the specification shall control. In describing embodiments of the present application, specific terminology is employed for the sake of clarity. However, the invention is not intended to be limited to the specific terminology so selected. Nothing in this specification should be considered as limiting the scope of the present invention. All examples presented are representative and non-limiting. The above-described embodiments may be modified or varied, without departing from the invention, as appreciated by those skilled in the art in light of the above teachings. It is therefore to be understood that, within the scope of the claims and their equivalents, the invention may be practiced otherwise than as specifically described.

What is claimed is:

1. A method of treating severe frostbite in a subject, comprising

- a) diluting about 1 mL of a pre-dilution composition comprising a concentrated solution of iloprost, said concentrated solution of iloprost comprising (i) about 0.1 mg/mL of iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, (ii) about 0.24 mg/mL of tromethamine, (iii) sodium chloride, and (iv) ethanol, in 0.9% sodium chloride, said diluted composition having a final concentration of about 1 µg/mL of iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof; and
 - b) administering the diluted composition to the subject by intravenous infusion, wherein the administration comprises a titration step on a first day of administration, said titration step being repeated on the second and third days of administration;
- wherein the administration on the first day comprises a starting dose and increments by which the dose is increased to a maximum dose, and wherein the admin-

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istration on the second and third days comprises the same starting dose and increments of increase as on the first day, but wherein the maximum dose achieved on each day may be different;

wherein the diluted composition is administered to the subject for a maximum of 8 consecutive days; and wherein the method reduces the risk of digit amputation in the subject when compared with standard of care treatments for severe frostbite.

2. The method of claim 1, wherein the administering is at one or more doses between about 0.25 ng/kg/min and about 2.0 ng/kg/min.

3. The method of claim 1, wherein the administering comprises a titration step on a first day of administration with a starting dose of about 0.25 ng/kg/min or about 0.5 ng/kg/min.

4. The method of claim 3, wherein the starting dose is about 0.5 ng/kg/min, and wherein the subject does not have a pre-existing Child-Pugh Class B or Child-Pugh Class C hepatic impairment.

5. The method of claim 3, wherein the starting dose is about 0.5 ng/kg/min and the dose is decreased to about 0.25 ng/kg/min at the beginning of the titration step when the subject has a renal impairment with eGFR less than 30 mL/min/m² and the subject cannot tolerate the starting dose of about 0.5 ng/kg/min.

6. The method of claim 3, wherein the starting dose is about 0.25 ng/kg/min, and wherein the subject has a Child-Pugh Class B or Child-Pugh Class C hepatic impairment.

7. The method of claim 3, wherein the dose is increased in increments of about 0.5 ng/kg/min every about 30 minutes during the titration step until a maximum dose is reached.

8. The method of claim 7, wherein the maximum dose is about 2.0 ng/kg/min or a highest dose the subject can tolerate between about 0.25 ng/kg/min and about 2.0 ng/kg/min.

9. The method of claim 3, wherein the titration step comprises decrease in the dose if the subject has a dose-limiting reaction.

10. The method of claim 1, wherein the diluted composition is administered continuously each day during a treatment period, wherein the continuous administration per day is for about 6 hours.

11. The method of claim 8, wherein the maximum dose or highest dose is maintained for the remainder of a treatment period after performing the titration step on the first day, the second day, and the third day.

12. The method of claim 1, wherein the severe frostbite is a stage 3 or a stage 4 frostbite.

13. The method of claim 1, wherein the pre-dilution composition is in a single dose vial in an amount of about 1 mL.

14. A method of treating severe frostbite in a subject, comprising

- a) diluting about 1 mL of a pre-dilution composition comprising a concentrated solution of iloprost, said concentrated solution of iloprost comprising (i) about 0.1 mg/mL of iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, (ii) about 0.24 mg/mL of tromethamine, (iii) sodium chloride, and (iv) about 8.1 mg/mL of ethanol, in 0.9% sodium chloride, said diluted composition having a final concentration of about 1 µg/mL of iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof; and
- b) administering the diluted composition to the subject by intravenous infusion, wherein the administration comprises a titration step on a first day of administration,

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said titration step being repeated on the second and third days of administration;

wherein the administration on the first day comprises a starting dose and increments by which the dose is increased to a maximum dose, and wherein the administration on the second and third days comprises the same starting dose and increments of increase as on the first day, but wherein the maximum dose achieved on each day may be different

wherein the diluted composition is administered to the subject for a maximum of 8 consecutive days; and wherein the method reduces the risk of digit amputation in the subject when compared with standard of care treatments for severe frostbite.

15. A method of treating severe frostbite in a subject comprising

- a) diluting about 1 mL of a pre-dilution composition comprising a concentrated solution of iloprost, said concentrated solution of iloprost comprising (i) about 0.1 mg/mL of iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, (ii) about 0.24 mg/mL of tromethamine, (iii) sodium chloride, and (iv) about 8.1 mg/mL of ethanol, in 0.9% sodium chloride, said diluted composition having a final concentration of about 1 µg/mL of iloprost, or a pharmaceutically acceptable salt or a stereoisomer thereof, wherein the composition has a pH of 8.3; and

- b) administering the diluted composition to the subject by intravenous infusion, wherein the administration comprises a titration step on a first day of administration, said titration step being repeated on the second and third days of administration;

wherein the administration on the first day comprises a starting dose and increments by which the dose is increased to a maximum dose, and wherein the administration on the second and third days comprises the same starting dose and increments of increase as on the first day, but wherein the maximum dose achieved on each day may be different;

wherein the diluted composition is administered to the subject for a maximum of 8 consecutive days; and wherein the method reduces the risk of digit amputation in the subject when compared with standard of care treatments for severe frostbite.

16. The method of claim 1, wherein the pre-dilution composition comprises about 0.242 mg/mL of tromethamine.

17. The method of claim 1, wherein the pre-dilution composition has a pH of about 8.3.

18. The method of claim 1, wherein the pre-dilution composition is formulated as a sterile solution.

19. The method of claim 1, wherein the pre-dilution composition comprises no preservatives.

20. The method of claim 1, wherein the diluted composition comprises about 100 mL of the 0.9% sodium chloride.

21. The method of claim 1, wherein the diluting is performed in an IV bag.

22. The method of claim 21, wherein the infusion bag comprises about 100 mL of the 0.9% sodium chloride.

23. The method of claim 21, wherein the infusion bag is made of polyvinyl chloride (PVC).

24. The method of claim 1, wherein the method reduces the risk of digit amputation by at least 15%, at least 20%, at least 25%, or at least 30% when compared with standard of care treatments for frostbite.

25. The method of claim **1**, wherein the method reduces the risk of digit amputation by about 35% to about 50% when compared with standard of care treatments for severe frostbite.

26. The method of claim **25**, wherein the method reduces the risk of digit amputation by about 40% when compared with standard of care treatments for severe frostbite.

27. The method of claim **12**, wherein the stage 3 frostbite is characterized by the presence of at least one digit of the subject having a lesion that extends just beyond a proximal phalanx.

28. The method of claim **12**, wherein the stage 4 frostbite is characterized by the presence of at least one digit of the subject having a lesion extending proximal to a metacarpal joint or a metatarsal joint.

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